AI POWERED KITCHEN ECOSYSTEM FOR FOOD WASTE REDUCTION AND NUTRITION OPTIMIZATION

Project Id: 25-26J-351

Final Individual Report

Shahmi MTM

IT22083982

BSc (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

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NUTRITIONAL GUIDANCE - PREDICTIVE HEALTH INTERVENTION SYSTEM

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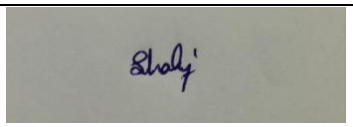
DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Shahmi MTM	IT22083982	

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

…….. ……

Signature of the supervisor Date

ABSTRACT

Nutritional imbalance and micronutrient deficiencies have become significant global health concerns, primarily due to poor dietary habits and lack of awareness regarding balanced nutrition. Traditional dietary assessment methods are often static, generalized, and incapable of adapting to individual intake patterns and health conditions. This research presents a Nutritional Guidance – Predictive Health Intervention System that leverages machine learning to analyze user dietary intake patterns, predict potential nutrient deficiencies, and support personalized dietary decision-making.

The proposed system integrates multiple structured datasets, including food nutritional composition, age-based nutrient requirements, and health-condition-based adjustments. A Gradient Boosting classification model is employed to predict deficiency risk levels categorized as LOW, MEDIUM, and HIGH based on user intake patterns and nutrient adequacy ratios (intake-to-requirement). Unlike conventional diet recommendation systems, the proposed system focuses on predictive intervention by identifying deficiencies before they become critical and providing actionable insights to improve nutritional balance.

Experimental results demonstrate that the model achieves an accuracy of 84.38% with a weighted F1-score of 0.8446 and stable cross-validation performance, indicating reliable predictive capability and generalization. The system enhances user awareness by identifying nutrient gaps and presenting interpretable risk levels, thereby supporting informed dietary decisions. This research contributes to intelligent health monitoring systems by combining predictive analytics, domain-driven nutritional evaluation, and user-centric guidance for preventive healthcare.

Keywords: *Nutritional Guidance, Machine Learning, Predictive Health, Deficiency Detection, Gradient Boosting, Nutrient Adequacy Ratios, Personalized Decision Support*

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LIST OF ABBREVIATIONS

- AI – Artificial Intelligence
- ML – Machine Learning
- API – Application Programming Interface
- IoT – Internet of Things
- NLP – Natural Language Processing
- PHIF – Predictive Health Intervention Framework
- ANPL – Adaptive Nutritional Personalization Layer
- SNRP – Smart Nutritional Recommendation Priority
- RF – Random Forest
- EDA – Exploratory Data Analysis
- RDA – Recommended Dietary Allowance
- CV – Cross-Validation

1. INTRODUCTION

Nutritional imbalance has emerged as a critical global health challenge, significantly affecting individual well-being, healthcare systems, and long-term quality of life. In recent years, the prevalence of micronutrient deficiencies and diet-related health conditions has increased due to changes in lifestyle, dietary habits, and food consumption patterns. Many individuals rely on irregular eating behaviors, processed food consumption, and nutritionally imbalanced diets, which contribute to deficiencies in essential nutrients such as protein, calcium, iron, and energy. These deficiencies can lead to serious health complications, including anemia, weakened immune systems, impaired growth, and chronic diseases. Despite the availability of dietary guidelines and nutritional recommendations, a large portion of the population remains unaware of their actual nutritional status and the potential risks associated with inadequate nutrient intake [1], [2].

One of the primary reasons for the persistence of nutritional deficiencies is the lack of effective and personalized dietary monitoring systems. Traditional methods of assessing nutrition, such as food diaries, manual calorie tracking, and general dietary guidelines, are often insufficient for capturing real-world eating behaviors. These approaches rely heavily on user consistency and accuracy, which can vary significantly among individuals. Furthermore, most conventional systems provide static recommendations that do not adapt to individual intake patterns, health conditions, or lifestyle differences. As a result, users are often unable to identify deficiencies early, leading to delayed intervention and increased health risks [3], [4].

Another critical limitation in existing nutritional systems is the absence of predictive capabilities. Most current solutions focus on retrospective analysis, evaluating past dietary intake without providing insights into future health risks. While some applications offer calorie tracking or basic nutrient summaries, they do not incorporate intelligent mechanisms to predict deficiency risks or provide proactive guidance. This lack of predictive intelligence limits their effectiveness as decision-support tools, as users are not guided toward preventive actions that can improve their nutritional health [5].

In addition, the complexity of nutritional requirements further complicates the problem. Nutritional needs vary significantly based on factors such as age, gender, physiological conditions, and underlying health issues. For instance, individuals with specific health conditions such as diabetes, hypertension, or anemia require customized dietary considerations that go beyond general nutritional guidelines. However, many existing systems fail to integrate such contextual factors effectively, resulting in generic outputs that may not be suitable for all users. This highlights the need for intelligent systems that can consider multiple dimensions of user data to provide accurate and personalized guidance [6].

Recent advancements in artificial intelligence and machine learning have opened new possibilities for addressing these challenges. Machine learning models are capable of analyzing structured data, identifying complex patterns, and generating predictive insights. In the context of nutritional guidance, these technologies can be used to analyze dietary intake patterns, compare them with recommended nutrient requirements, and predict potential deficiencies. The proposed system leverages nutrient adequacy ratios (intake-to-requirement) as key features, enabling normalization across diverse user profiles and improving the accuracy of deficiency risk prediction. Unlike traditional methods, such models provide interpretable and data-driven insights that support better decision-making [7], [8].

Several studies have demonstrated the effectiveness of machine learning in healthcare and nutritional analysis. Classification algorithms such as Random Forest, Support Vector Machines, and Neural Networks have been successfully applied to predict health conditions and assess risk levels based on input data. These models are particularly suitable for nutritional applications, where multiple variables interact in complex and non-linear ways. However, existing research often focuses on disease detection rather than early-stage deficiency prediction, and many systems lack integration with real-world dietary intake data and user-specific requirements [9], [10].

Motivated by these limitations, this research proposes a Nutritional Guidance – Predictive Health Intervention System that combines machine learning with structured nutritional datasets to provide personalized and proactive dietary guidance. The system is designed to analyze user intake patterns, evaluate nutrient consumption, and predict deficiency risk levels

using a Gradient Boosting classification model. Unlike conventional diet recommendation systems, the proposed approach focuses on predictive intervention by identifying potential deficiencies before they become critical and supporting informed dietary decisions.

The system integrates multiple data sources, including food nutritional composition data, age-based nutrient requirements, and health-condition-based adjustments. By comparing actual intake with recommended values through nutrient adequacy ratios, the system identifies gaps in nutrient consumption and classifies the level of deficiency risk. Based on these predictions, it provides guidance aimed at improving nutritional balance and supporting overall health.

The significance of this research lies in its ability to bridge the gap between nutritional data analysis and practical health intervention. Rather than simply tracking dietary intake, the proposed system transforms raw data into actionable insights that guide users toward healthier eating behaviors. By combining predictive analytics, domain-driven nutritional evaluation, and user-centric guidance, the system provides a comprehensive solution for addressing nutritional deficiencies in real-world settings.

Overall, this study contributes to the field of intelligent health systems by introducing a predictive and user-centric approach to nutritional guidance. The proposed system provides reliable deficiency risk prediction with interpretable outputs and enhances usability by presenting meaningful insights that users can easily understand and apply. In doing so, it supports preventive healthcare strategies and promotes long-term nutritional well-being.

1.1 Background & Literature Survey

Understanding nutritional health and dietary behavior requires a multidisciplinary perspective that combines nutrition science, healthcare analytics, and computational modeling. Nutritional imbalance is a gradual and often unnoticed process driven by inadequate intake of essential nutrients over time. Unlike acute diseases, nutritional deficiencies develop progressively through repeated dietary patterns that fail to meet the body's physiological requirements. Factors such as poor dietary choices, irregular eating habits, socioeconomic conditions, and lack of nutritional awareness significantly contribute to this issue. Deficiencies in nutrients such as iron, protein, calcium, and energy can lead to serious health consequences, including anemia, weakened immunity, impaired cognitive development, and chronic illnesses. These effects highlight the importance of continuous monitoring and early detection of nutritional imbalance rather than relying solely on symptom-based diagnosis [11].

Traditional approaches to nutritional assessment have primarily relied on dietary surveys, food frequency questionnaires, and manual food diaries. These methods are widely used in clinical and research settings due to their simplicity and ability to capture general eating patterns. However, they are highly dependent on user accuracy and consistency, which often leads to unreliable data. Moreover, these approaches provide only retrospective insights into dietary behavior and lack the ability to offer predictive or real-time guidance. Nutritional recommendations derived from such methods are typically generalized and do not adapt to individual variability in dietary habits, health conditions, or metabolic requirements. As a result, traditional systems are often insufficient for providing effective and personalized nutritional guidance in real-world scenarios [12].

To address some of these limitations, digital health technologies and mobile applications have been introduced for dietary tracking and calorie monitoring. These systems enable users to log food intake and receive basic nutritional summaries, improving awareness of dietary habits. However, their functionality is largely limited to descriptive analytics rather than predictive intelligence. While they can display nutrient intake levels, they do not provide insights into future health risks or identify potential deficiencies before they become critical. This limitation

restricts their usefulness in preventive healthcare, where early intervention is essential for maintaining optimal health [13].

Recent advancements in artificial intelligence and machine learning have significantly transformed the field of healthcare analytics, including nutritional assessment. Machine learning models are capable of analyzing structured data, identifying complex patterns, and generating predictive insights that support decision-making. In the context of nutrition, these models can be used to analyze dietary intake patterns, compare them with recommended nutrient requirements, and predict the likelihood of deficiencies. In particular, the use of nutrient adequacy ratios (intake-to-requirement) enables normalization across different user profiles, improving the reliability and interpretability of deficiency risk prediction. Unlike traditional statistical approaches, machine learning techniques can effectively model non-linear relationships between multiple nutritional and demographic variables [14].

Various machine learning algorithms have been applied to nutritional and health prediction tasks, including decision trees, support vector machines, neural networks, and ensemble methods such as Random Forest and Gradient Boosting. Ensemble learning methods are particularly effective due to their ability to combine multiple weak learners and improve predictive performance. Gradient Boosting, in particular, has demonstrated strong performance in structured tabular datasets by sequentially minimizing prediction errors and capturing complex relationships between features. These properties make it suitable for nutritional deficiency risk prediction tasks where multiple interacting variables influence outcomes [15].

In addition to predictive modeling, recent research has explored the integration of nutritional data with personalized guidance systems. These systems aim to support dietary improvements by analyzing individual intake patterns and identifying nutritional gaps. However, many existing systems focus primarily on recommendation without incorporating predictive mechanisms that identify underlying deficiencies. This separation between prediction and guidance limits their effectiveness, as outputs may not always be based on an accurate assessment of nutritional risk [16].

Another emerging trend in nutritional research is the use of integrated datasets that combine food composition data, demographic information, and health condition adjustments. Such datasets enable more comprehensive analysis of nutritional requirements and allow systems to account for variability among individuals. For instance, nutrient requirements differ across age groups and are influenced by medical conditions such as diabetes, hypertension, or anemia. Incorporating these factors into predictive models enhances their accuracy and relevance, enabling more personalized and context-aware nutritional analysis [17].

Despite these advancements, several challenges remain in the development of intelligent nutritional guidance systems. One major limitation is the lack of systems that simultaneously integrate intake pattern analysis, predictive modeling, and user-centric guidance within a unified framework. Many existing approaches focus on only one aspect, such as tracking or recommendation, without providing a complete solution for nutritional intervention. Additionally, the interpretability of machine learning models remains a concern, as users may not fully understand how predictions are generated, which can reduce trust and adoption. Furthermore, data quality and user input accuracy continue to influence system performance, highlighting the need for robust preprocessing and validation mechanisms [18].

A critical research gap therefore exists in designing systems that not only analyze dietary intake but also predict deficiency risks and provide actionable guidance in a personalized manner. Such systems must combine predictive intelligence with domain knowledge and user-specific factors to deliver meaningful insights. This requires the integration of machine learning techniques with structured nutritional datasets and decision-support mechanisms that guide users toward improved dietary behavior.

In summary, the literature reveals a transition from traditional dietary assessment methods toward intelligent, data-driven nutritional systems. While conventional approaches provide foundational understanding, they lack adaptability and predictive capability. Machine learning-based systems address many of these limitations by enabling dynamic and personalized analysis, yet challenges related to integration, usability, and interpretability remain. These limitations motivate the development of advanced predictive health

intervention systems, such as the one proposed in this study, which combines intake pattern analysis, deficiency prediction, and user-centered guidance within a unified framework.

1.2 Research Gap

Despite significant advancements in nutritional analysis, dietary tracking systems, and machine learning-based healthcare applications, several critical limitations persist in existing nutritional guidance solutions. Most current systems focus primarily on descriptive analytics, such as calorie tracking or nutrient summarization, without incorporating predictive intelligence. Traditional nutritional assessment approaches rely on static dietary guidelines and retrospective data analysis, which do not capture the dynamic nature of real-world dietary behavior. While these methods provide a general understanding of nutrient intake, they fail to identify emerging deficiencies or provide early warnings that support preventive healthcare. As a result, users often remain unaware of nutritional imbalances until symptoms begin to appear, limiting the effectiveness of early intervention strategies [19], [20].

Another major limitation in existing systems is the lack of integration between multiple data sources required for accurate nutritional assessment. Many applications consider only food intake data without incorporating age-based nutrient requirements, health-condition-specific adjustments, or individual variability. Nutritional needs differ significantly across individuals based on demographic and physiological factors, and failing to account for these variations leads to generalized and often inaccurate outputs. Although some research studies have attempted to integrate multiple datasets, these approaches are often limited in scope and do not provide a unified framework for comprehensive nutritional analysis [21].

Furthermore, while machine learning has been increasingly applied in healthcare prediction systems, its application in nutritional guidance remains limited and fragmented. Existing machine learning models are typically designed for disease prediction rather than early-stage deficiency detection. These models often focus on identifying already-developed health conditions rather than predicting potential risks based on dietary intake patterns. Additionally, many systems operate as standalone prediction models without integrating user-centered guidance that translates predictions into actionable insights. This separation between prediction and guidance reduces the practical usefulness of these systems, as users are not effectively supported in improving their nutritional status [22], [23].

Another critical gap lies in the absence of personalized and context-aware mechanisms in nutritional systems. Most existing solutions generate uniform outputs that do not fully reflect user-specific factors such as intake patterns, demographic characteristics, and nutritional requirements. In real-world scenarios, dietary habits are highly dynamic and influenced by lifestyle, cultural preferences, and accessibility of food. However, current systems rarely incorporate structured feature engineering approaches, such as nutrient adequacy ratios (intake-to-requirement), which are essential for normalizing intake data across diverse users. The absence of such mechanisms limits the accuracy and interpretability of deficiency detection.

In addition, many existing systems lack decision-support capabilities that help users translate nutritional insights into practical actions. Even when nutrient deficiencies are identified, users are often not provided with clear and interpretable insights to guide their decisions. Furthermore, interpretability remains a challenge in many machine learning-based systems, as complex models do not always provide understandable outputs. This lack of transparency can reduce user trust and hinder adoption in real-world applications [25].

Therefore, there is a clear need for an intelligent and integrated nutritional guidance system that combines intake pattern analysis, predictive modeling, and user-centered decision support within a unified framework. Such a system should not only analyze dietary intake but also predict deficiency risks proactively, incorporate user-specific factors such as age and health conditions, and provide interpretable insights that support informed dietary decisions. Additionally, the system should ensure usability and clarity, enabling users to understand and act upon the outputs effectively.

This research addresses these gaps by proposing a Nutritional Guidance – Predictive Health Intervention System that integrates machine learning with structured nutritional datasets to deliver predictive, personalized, and interpretable dietary guidance. The system combines intake-based analysis, nutrient adequacy ratio-based feature engineering, and deficiency risk prediction using a Gradient Boosting model within a cohesive framework. This approach enables early detection of nutritional imbalances and supports preventive healthcare strategies through informed decision-making.

Table 1.1 – Research Gap Table

System / Study	Approach Used	Key Strength	Limitation	Gap Identified	Your System Contribution
WHO Nutrition Guidelines [19]	Static dietary recommendations	Widely accepted standards	Not personalized	No intake-based adaptation	Personalized intake-based prediction
Dietary Tracking Apps [20]	Calorie & nutrient logging	Easy to use	No prediction capability	No risk detection	ML-based deficiency prediction
ML Healthcare Models [22]	Disease prediction models	High accuracy	Focus on diseases only	No early deficiency prediction	Predictive deficiency classification
Nutrition Recommendation Systems [23]	Food recommendation algorithms	Personalized suggestions	Not linked to deficiency prediction	Weak connection to actual nutrient gaps	Insight-driven dietary guidance based on deficiency risk
Integrated Nutrition Studies [21]	Multi-dataset analysis	Better accuracy	Limited integration	No unified framework	Integrated multi-source nutritional analysis
AI Health Systems [25]	Complex ML models	Advanced analytics	Low interpretability	Lack of user trust	Interpretable risk prediction outputs
Existing Systems Overall	Fragmented approaches	Partial solutions	Lack of integration	No unified predictive framework	Hybrid predictive health intervention system

1.3 Research Problem

The increasing prevalence of nutritional deficiencies and diet-related health issues highlights a critical need for more intelligent and context-aware systems for dietary assessment and intervention. Despite the availability of general dietary guidelines and nutritional tracking tools, existing approaches remain insufficient for real-world decision-making due to their reliance on static recommendations and limited analytical capabilities. Conventional nutritional assessment methods, including food diaries, dietary surveys, and guideline-based evaluation, assume generalized intake patterns and fail to capture the variability of individual eating behaviors. These methods often depend on user consistency and accuracy, which can lead to incomplete or unreliable data. As a result, they are unable to provide timely insights into nutritional imbalances, leading to delayed identification of deficiencies and reduced effectiveness of preventive interventions [19], [20].

Although modern digital applications and dietary tracking systems have improved user engagement by enabling real-time food logging and nutrient visualization, they primarily function as descriptive tools rather than predictive systems. Most existing solutions provide retrospective summaries of nutrient intake without assessing the potential health risks associated with long-term dietary patterns. Consequently, users are not informed about emerging deficiencies or future health implications, limiting the role of these systems in proactive health management. Furthermore, these applications often provide generalized feedback that does not account for individual variability in age, health conditions, and nutritional requirements, thereby reducing the accuracy and relevance of their outputs [21], [23].

Recent advancements in machine learning have demonstrated significant potential in healthcare prediction and data-driven decision-making. However, the application of these techniques in nutritional guidance remains limited and fragmented. Many existing machine learning models are designed for disease detection or classification based on clinical data rather than dietary intake patterns. These models typically focus on identifying already developed health conditions rather than predicting early-stage nutritional deficiencies. In addition, most systems operate as standalone predictive models without integrating user-

centered decision-support mechanisms that translate predictions into actionable insights. This lack of integration reduces their practical usefulness, as users are not effectively supported in improving their nutritional intake based on identified risks [22], [25].

Another critical limitation in current nutritional systems is the lack of structured feature representation that accurately captures nutritional adequacy. Most existing approaches rely on raw intake values or generalized thresholds without normalizing intake based on individual requirements. In real-world scenarios, nutrient intake must be interpreted relative to personalized requirements to accurately identify deficiencies. The absence of mechanisms such as nutrient adequacy ratios (intake-to-requirement) limits the effectiveness of deficiency prediction models and reduces interpretability across diverse user groups.

Moreover, existing nutritional guidance systems lack comprehensive decision-support capabilities that assist users in making informed dietary choices. Even when nutrient deficiencies are identified, users are often not provided with clear and interpretable insights that directly relate to their intake patterns. The absence of an integrated framework that combines intake analysis, deficiency prediction, and user-centered guidance results in fragmented solutions that fail to deliver practical value. Furthermore, interpretability remains a challenge in many machine learning-based systems, as complex models do not always provide transparent explanations for their predictions, reducing user trust and limiting adoption in real-world applications [25].

Therefore, the core research problem addressed in this study is the development of a predictive and personalized nutritional guidance system that can effectively analyze dietary intake patterns and support preventive health intervention. Specifically, the challenge lies in designing a system that evaluates user nutrient intake, transforms it into meaningful representations such as nutrient adequacy ratios, and predicts deficiency risks using machine learning techniques. The system must also incorporate age-based and health-condition-aware requirements while providing interpretable outputs that support informed dietary decision-making. Addressing this problem requires the integration of data-driven predictive modeling with domain-specific nutritional knowledge within a unified and practical framework. The inability of existing systems to simultaneously achieve prediction accuracy, personalization,

interpretability, and usability underscores the necessity for the proposed approach in this research.

2. OBJECTIVES

2.1 Main Objectives

The primary objective of this research is to design and develop an intelligent Nutritional Guidance – Predictive Health Intervention System that analyzes dietary intake patterns, predicts potential nutrient deficiencies, and supports informed dietary decision-making. The proposed system integrates machine learning-based prediction with structured nutritional datasets, including food composition data, age-based nutrient requirements, and health-condition-aware adjustments, to address the limitations of traditional dietary assessment approaches.

In addition to improving the reliability of nutritional analysis, the system is designed to deliver context-aware and user-specific insights by evaluating individual intake behavior rather than relying on generalized dietary recommendations. By combining predictive intelligence with domain-driven nutritional evaluation, the research aims to provide a comprehensive solution that identifies nutritional imbalances at an early stage and supports preventive healthcare through interpretable outputs.

Furthermore, the system enhances user awareness of nutrient intake gaps and promotes informed dietary decisions by presenting deficiency risk levels in a clear and understandable manner. Rather than directly prescribing meal plans, the system functions as a decision-support tool that guides users toward improving their nutritional balance based on identified deficiencies.

2.2 Specific Objectives

- To develop a machine learning-based classification model capable of predicting nutrient deficiency risk levels (LOW, MEDIUM, HIGH) based on user dietary intake patterns and nutrient adequacy ratios.
- To design and implement a data processing pipeline that aggregates user food intake, maps nutritional values from food composition datasets, and compares them with recommended nutrient requirements.
- To integrate age-based nutrient requirement datasets and health-condition-aware adjustments to ensure that predictions are personalized and context-aware.
- To implement feature engineering techniques, including nutrient adequacy ratios (intake-to-requirement), to improve the accuracy and interpretability of deficiency risk prediction.
- To design a system workflow that connects intake analysis and deficiency risk prediction within a unified framework for seamless user experience.
- To evaluate the performance of the machine learning model using appropriate metrics such as accuracy, precision, recall, weighted F1-score, and cross-validation, ensuring reliability and generalization capability.
- To assess the practical usability of the system in terms of user experience, interpretability, and effectiveness in supporting informed dietary decision-making.
- To analyze the impact of the system in promoting preventive healthcare by enabling early detection of nutritional deficiencies and improving user awareness of nutrient intake gaps.

3. METHODOLOGY

3.1 System Overview

The proposed Nutritional Guidance - Predictive Health Intervention System is designed as a hybrid intelligent framework that integrates dietary intake analysis, machine learning-based prediction, and structured nutritional knowledge to provide user-centered health insights. Unlike traditional nutritional systems that rely on static recommendations or manual tracking, the proposed system functions as a predictive decision-support platform that identifies potential nutrient deficiencies and supports informed dietary decision-making.

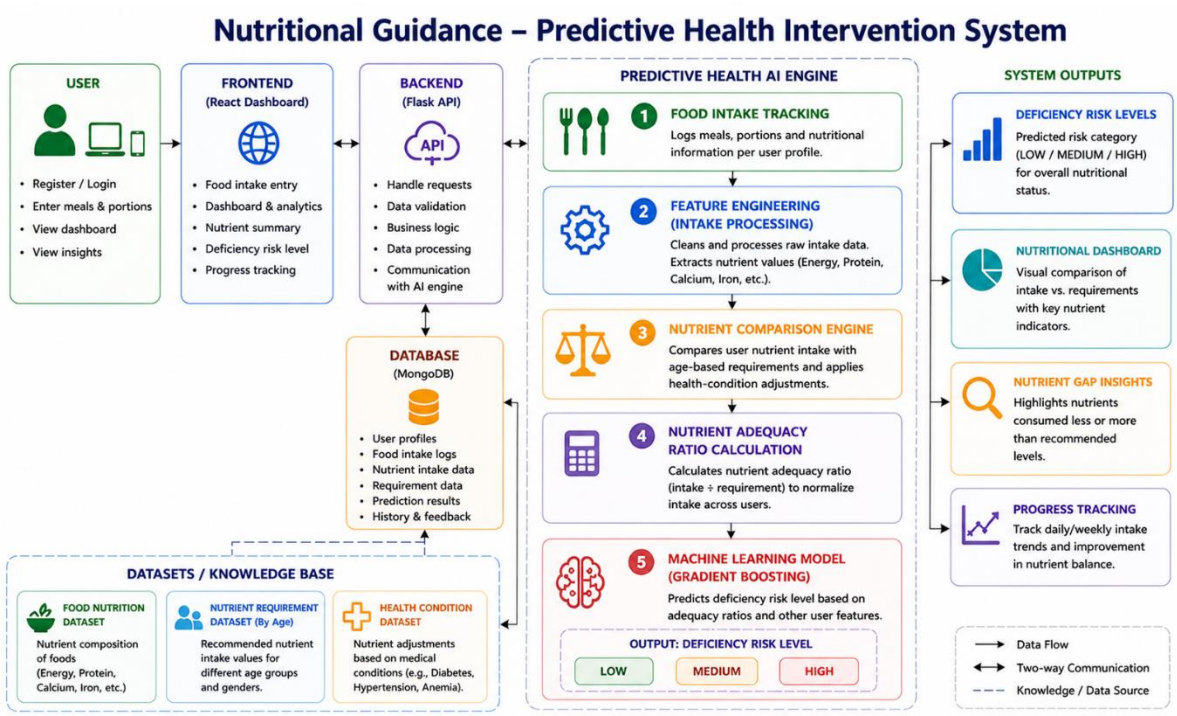

Figure 3.1 – System Architecture

As illustrated in Fig. 3.1, the system architecture follows a modular pipeline in which user inputs are processed through multiple analytical components before generating outputs in the form of deficiency risk levels and nutritional insights. The system is designed to ensure a seamless flow of data from intake collection to predictive analysis while maintaining interpretability and usability.

The workflow begins with user interaction through a web-based interface, where dietary intake information such as meals, portion sizes, and food items is recorded. These inputs represent the user's daily consumption patterns and form the foundational data for nutritional analysis. The frontend communicates with the backend through RESTful APIs, where the collected data is validated and processed. User profiles, intake logs, and historical data are stored in a MongoDB database, enabling efficient storage and retrieval for analysis.

Once intake data is collected, it is passed to the data processing module, where raw inputs are cleaned, structured, and transformed into meaningful nutritional features. This includes aggregating nutrient values such as energy, protein, calcium, and iron from the food composition dataset. These values represent the user's actual nutrient intake over a selected period.

Following this, the system performs requirement matching using age-based and health-condition-aware nutritional datasets. In this stage, the user's intake is compared against recommended nutrient requirements. To improve interpretability and model performance, the system computes nutrient adequacy ratios (intake-to-requirement), which normalize intake across different users and provide a consistent basis for deficiency detection.

The processed feature set, including user demographics, intake values, requirement values, and adequacy ratios, is then passed to the machine learning component. The system utilizes a Gradient Boosting classification model to predict deficiency risk levels categorized as LOW, MEDIUM, or HIGH. The model captures relationships between nutrient intake patterns and deficiency risk, enabling early identification of nutritional imbalances.

The prediction results are forwarded to the output layer, where users receive interpretable feedback, including deficiency risk levels and nutrient intake summaries. The system presents these insights through dashboards and reports, allowing users to understand their nutritional status and identify gaps in their diet. Rather than directly prescribing meal plans, the system supports decision-making by highlighting deficiencies and guiding users toward improving their nutritional balance.

Overall, the system architecture demonstrates the integration of intake data acquisition, feature engineering, requirement-based evaluation, and machine learning-based prediction within a unified framework. By combining predictive analytics with domain-driven nutritional evaluation, the system provides reliable and interpretable insights that support preventive healthcare and improved dietary awareness.

3.2 System Workflow

The system workflow of the proposed Nutritional Guidance – Predictive Health Intervention System is designed to ensure a structured, intelligent, and user-centric process that transforms raw dietary intake data into meaningful nutritional insights. The workflow follows a sequential pipeline, beginning with user interaction and progressing through multiple analytical stages within the predictive health AI engine before generating final outputs.

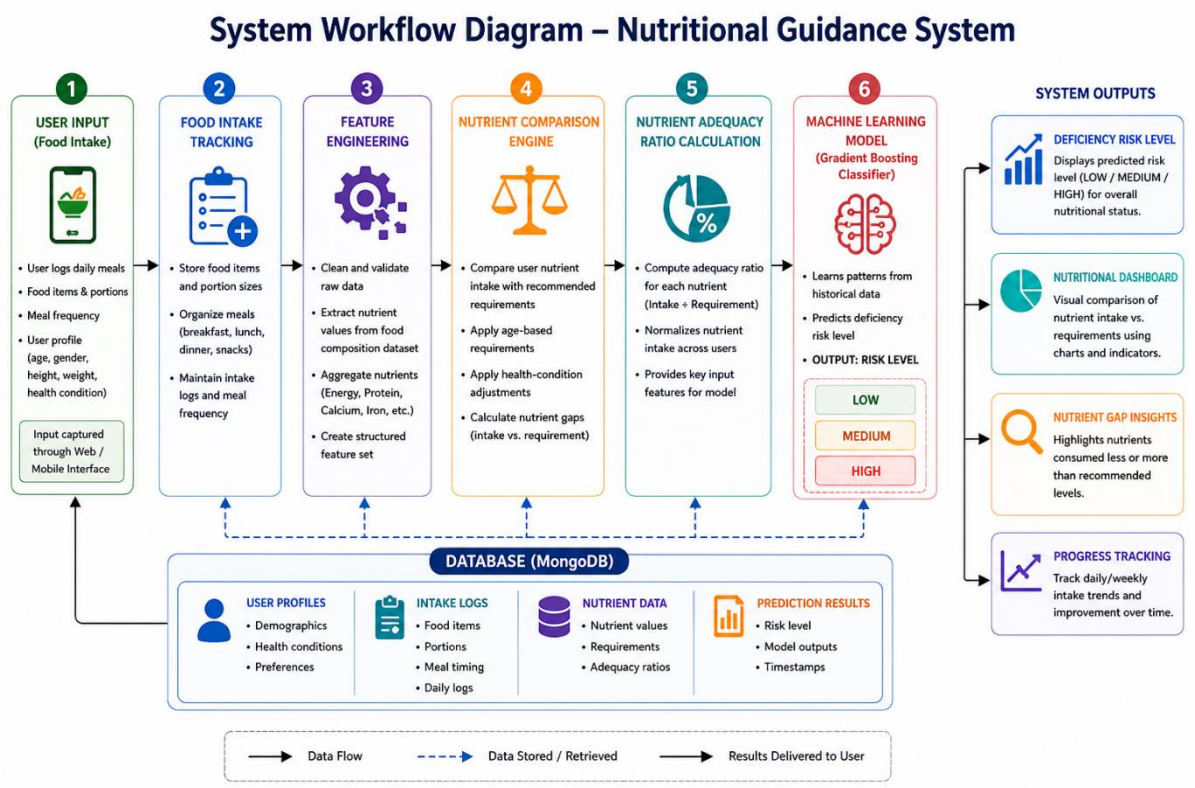

Figure 3.2 – System Workflow Diagram

As illustrated in Fig. 3.2, the system operates as a structured data-processing pipeline in which user inputs, feature engineering, and predictive modeling are integrated to deliver accurate and interpretable deficiency risk insights. The workflow is designed to support preventive healthcare by identifying potential nutritional imbalances at an early stage and assisting users in making informed dietary decisions.

The workflow begins with the food intake tracking stage, where users input their daily dietary consumption through the system interface. This includes details such as food items consumed, portion sizes, and meal frequency. These inputs form the foundational dataset representing the user's intake patterns. The system captures intake data consistently to provide a realistic representation of user behavior over time.

Once the intake data is collected, it is passed to the feature engineering stage, where raw inputs are transformed into structured features suitable for analysis. This stage involves data cleaning, normalization, and aggregation of nutritional values. Nutrient information such as energy, protein, calcium, and iron is extracted from the food composition dataset and aggregated based on user intake.

Following feature engineering, the structured data is forwarded to the nutrient comparison engine, which compares the user's nutrient intake against recommended nutrient requirements derived from age-based standards and health-condition-aware adjustments. This comparison allows the system to identify gaps between actual intake and required levels.

To improve prediction accuracy and consistency across different users, the system computes nutrient adequacy ratios (intake-to-requirement). These ratios provide a normalized representation of nutrient intake and serve as key input features for the machine learning model.

The processed feature set, including intake values, requirement values, and adequacy ratios, is then passed to the machine learning prediction module. The system utilizes a Gradient Boosting classification model to predict deficiency risk levels categorized as LOW, MEDIUM, or HIGH. The model captures relationships between nutrient intake patterns and deficiency risk, enabling early identification of nutritional imbalances.

Following prediction, the system generates outputs through the user interface. These outputs include deficiency risk levels, nutritional summaries, and visual analytics presented through a dashboard. The dashboard provides insights into nutrient intake versus recommended values, allowing users to understand their nutritional status and identify gaps in their diet. The system

functions as a decision-support tool by presenting interpretable results that guide users toward improving their nutritional balance.

All processed data, including intake records and prediction results, are stored in the system database. This enables historical tracking and supports future analysis of user intake patterns. The stored data contributes to maintaining system functionality and allows users to monitor their progress over time.

Overall, the system workflow demonstrates a seamless integration of data collection, feature engineering, requirement-based evaluation, and machine learning-based prediction within a unified framework. By combining predictive analytics with domain-driven nutritional analysis, the proposed workflow provides reliable and interpretable insights that support preventive healthcare and improved dietary awareness.

3.3 Data Featuring

The effectiveness of the proposed Nutritional Guidance – Predictive Health Intervention System is fundamentally dependent on the quality, diversity, and relevance of the data used for analysis and model training. In this study, multiple structured tabular datasets are utilized to capture the relationships between dietary intake patterns, nutrient composition, and recommended nutritional requirements. These datasets collectively enable the system to analyze user consumption behavior, identify nutritional imbalances, and support accurate deficiency risk prediction.

The primary dataset used in this system is the food nutrition dataset, which contains detailed nutritional composition information for a wide range of food items. This dataset includes essential nutrient values such as energy, protein, calcium, and iron, which are critical indicators for assessing an individual's nutritional status. Each food item is associated with its corresponding nutrient values per standard serving size, allowing the system to calculate total nutrient intake based on user consumption. This dataset serves as the foundation for transforming raw dietary input into structured nutritional features.

In addition to the food nutrition dataset, the system incorporates a nutrient requirement dataset that defines recommended daily intake values for different age groups and demographic categories. This dataset provides benchmark values against which user intake is evaluated. By integrating age-based requirements, the system ensures that nutritional assessment is personalized and context-aware, rather than relying on generalized standards.

Furthermore, a health-condition adjustment dataset is integrated to enhance personalization. This dataset includes modifications to nutrient requirements based on specific health conditions such as diabetes, hypertension, and anemia. By incorporating these adjustments, the system accounts for user-specific health factors, improving the relevance and accuracy of deficiency risk prediction.

The combined dataset structure includes both categorical and numerical features that describe user intake and nutritional requirements. Core input features include user demographic attributes (age, gender, health condition), nutrient intake values (energy, protein, calcium,

iron), and corresponding requirement values. In addition to these, derived features such as nutrient adequacy ratios (intake-to-requirement) are computed during preprocessing. These ratios represent the most influential features in the model, as they normalize nutrient intake across different users and directly capture the gap between actual consumption and required levels.

The final feature set consists of a structured combination of user profile features, intake features, requirement features, and ratio-based features, enabling the model to learn meaningful relationships between dietary behavior and nutritional risk. This feature engineering approach ensures that the model is both interpretable and robust, as it incorporates domain-driven nutritional knowledge into the learning process.

The target variable used in this study is the nutrient deficiency risk level, categorized into three classes: LOW, MEDIUM, and HIGH. This classification is derived from the relationship between nutrient intake and requirement levels and serves as an interpretable representation of nutritional status. By formulating the problem as a classification task, the system provides clear and user-friendly outputs that support practical decision-making.

Overall, the integration of multiple structured datasets and the use of nutrient adequacy ratios as key engineered features provide a strong data foundation for predictive modeling. This data-driven approach enables the machine learning model to generate reliable deficiency risk predictions while maintaining interpretability and consistency across diverse user profiles.

3.4 Data preprocessing

Data preprocessing is a critical stage in the proposed Nutritional Guidance – Predictive Health Intervention System, as it ensures the accuracy, consistency, and reliability of the data before it is used for feature engineering and machine learning prediction. Since the system integrates multiple datasets, including food nutrition data, nutrient requirement standards, and health-condition-based adjustments, a structured preprocessing pipeline is essential to prepare the data for effective analysis and modeling.

The initial step in the preprocessing pipeline involves data cleaning, where inconsistencies, duplicate entries, and irrelevant records are identified and removed. Food nutrition datasets may contain duplicate food items, inconsistent naming conventions, or incomplete records, which can negatively affect the accuracy of nutrient calculations. Therefore, duplicate entries are eliminated, and food names are standardized to ensure consistency across all datasets. Additionally, unnecessary attributes that do not contribute to the prediction process are removed to reduce noise and improve computational efficiency.

Handling missing values is another important component of the preprocessing stage. Missing nutrient values can lead to inaccurate intake calculations, which directly impacts deficiency prediction. Therefore, records with critical missing values are either removed or handled using appropriate imputation strategies. Numerical features such as nutrient values (energy, protein, calcium, and iron) are filled using mean values or domain-informed defaults when necessary, while categorical attributes are standardized to maintain consistency.

Following data cleaning, the preprocessing pipeline performs data integration, where multiple datasets are combined into a unified structure. The food nutrition dataset is mapped with user intake data to calculate total nutrient consumption, while the nutrient requirement dataset provides benchmark values for comparison. Additionally, the health-condition adjustment dataset is integrated to modify nutrient requirements based on user-specific conditions. This integration ensures that the system performs personalized and context-aware nutritional analysis.

Feature engineering is then applied to convert raw intake data into meaningful analytical features. Nutrient values from individual food items are aggregated based on user intake, resulting in total nutrient consumption over a selected period. In addition to raw intake values, derived features such as nutrient adequacy ratios (intake-to-requirement) are computed. These ratios play a crucial role in the system, as they normalize nutrient intake across users and directly represent the gap between actual consumption and required levels. As a result, they serve as the most influential features for deficiency risk prediction.

To ensure compatibility with the machine learning model, relevant features are selected and structured into a consistent numerical format. The final feature set includes user demographic features (age, gender, health condition), nutrient intake values, requirement values, and ratio-based features. Since the model primarily operates on numerical inputs, categorical encoding requirements are minimal, and emphasis is placed on maintaining a clean and well-structured feature matrix.

Finally, the processed data is validated before being passed to the machine learning model. This validation step checks for anomalies, inconsistencies, or extreme outliers that may affect prediction performance. Ensuring data quality at this stage is essential for maintaining model reliability and stability.

Overall, the data preprocessing stage transforms raw dietary and nutritional data into a high-quality, structured format suitable for predictive modeling. By addressing data quality, integration, and feature representation, this stage significantly enhances the system's ability to generate reliable deficiency risk predictions and provide interpretable nutritional insights.

3.5 Feature Engineering

Feature engineering plays a critical role in enhancing the predictive capability of the proposed Nutritional Guidance – Predictive Health Intervention System by transforming raw dietary intake data into meaningful representations that can be effectively utilized by the machine learning model. Given that the system integrates multiple structured datasets, including food nutrition data, user intake logs, nutrient requirements, and health-condition adjustments, a systematic feature engineering process is implemented to capture the relationships between dietary patterns and nutritional health outcomes.

One of the primary steps in feature engineering involves the transformation of raw intake data into aggregated nutrient features. Individual food items consumed by the user are mapped to their corresponding nutritional values using the food nutrition dataset. These values are then aggregated based on portion sizes and meal frequency to compute total nutrient intake over a selected period. Key nutrients such as energy, protein, calcium, and iron are selected as primary indicators, as they are essential for assessing overall nutritional status and deficiency risk. This aggregation process converts fragmented intake data into a structured representation that reflects the user's actual consumption behavior.

In addition to raw nutrient values, derived features are generated to improve the model's ability to detect nutritional imbalances. A key derived feature in this system is the nutrient adequacy ratio, calculated as the proportion of actual intake to recommended nutrient requirements (intake-to-requirement). These ratios provide a normalized representation of nutrient consumption, allowing the system to evaluate intake relative to personalized requirements. As a result, adequacy ratios directly capture the gap between actual consumption and required levels and serve as the most influential features for deficiency risk prediction.

Another important aspect of feature engineering is the integration of contextual and domain-specific information. Nutrient requirement values are incorporated from the age-based dataset to ensure that intake comparisons are personalized according to user demographics. Additionally, adjustments from the health-condition dataset are applied to modify nutrient requirements for users with specific medical conditions such as diabetes, hypertension, or

anemia. These contextual features enable the system to perform personalized and context-aware analysis.

The final feature set is composed of a structured combination of user profile features, intake features, requirement features, and ratio-based features. Specifically, the model utilizes features such as age, gender (encoded), and health-condition indicators, along with nutrient intake values (energy, protein, calcium, iron), corresponding requirement values, and adequacy ratios for each nutrient. This results in a compact yet informative feature space that effectively captures the relationship between dietary intake and nutritional risk.

Since the proposed system utilizes a Gradient Boosting model, which operates effectively on structured numerical data, minimal categorical encoding is required. User attributes such as gender and health conditions are encoded into numerical or binary representations where necessary. Unlike traditional high-dimensional datasets, this approach focuses on domain-relevant features, improving both interpretability and computational efficiency.

Feature scaling and normalization are also considered to ensure consistency across input variables. Although tree-based models such as Gradient Boosting are generally less sensitive to feature scaling, maintaining consistent feature ranges improves model stability and interpretability. Nutrient values and derived ratios are normalized where necessary to prevent disproportionate influence from features with larger magnitudes.

Overall, the feature engineering process enhances the representational power of the dataset by combining nutrient aggregation, adequacy ratio-based feature construction, and contextual personalization. This approach ensures that the machine learning model receives informative and relevant inputs, enabling accurate prediction of deficiency risk levels. By transforming raw dietary data into structured and meaningful features, this stage plays a crucial role in the effectiveness of the proposed predictive health intervention system.

3.6 Predictive Health Intervention Framework

3.6.1 Conceptual Foundation and Motivation

The Predictive Health Intervention Framework (PHIF) is introduced as the core methodological contribution of this research, designed to address the fundamental limitations observed in existing nutritional guidance and dietary assessment systems. Unlike conventional approaches that focus solely on static dietary recommendations or basic intake tracking, the proposed framework integrates predictive modeling, domain-specific nutritional knowledge, and user-centered analysis to provide a comprehensive and proactive health assessment system.

Current nutritional systems generally fall into two categories: guideline-based systems and data-driven machine learning-based systems. Rule-based nutritional systems rely on generalized dietary guidelines derived from population-level studies. While these guidelines provide a foundational understanding of nutritional requirements, they do not account for individual variability in dietary habits, health conditions, and lifestyle factors. As a result, such systems offer static outputs that are often insufficient for real-world applications, where personalized and context-aware analysis is required [19], [21].

On the other hand, machine learning-based approaches have demonstrated the ability to analyze complex relationships between dietary intake and health outcomes. These systems leverage data-driven techniques to identify patterns and predict potential risks. However, many existing models operate as standalone predictive tools without incorporating domain-specific nutritional constraints or interpretable outputs. Consequently, while predictions may be statistically accurate, they often lack practical usability and fail to provide meaningful insights for users to understand and improve their dietary behavior [22], [25].

This gap between predictive capability and practical usability highlights the need for an integrated framework that combines both paradigms. The design of PHIF is therefore inspired by principles of hybrid intelligent systems, where multiple reasoning mechanisms are combined to enhance robustness, interpretability, and decision-support capability. In the context of nutritional guidance, such integration ensures that predictions are not only data-

driven but also aligned with established nutritional standards and real-world usability requirements.

Furthermore, the integration of structured nutritional knowledge within a predictive framework aligns with research in intelligent decision-support systems, where the balance between accuracy, interpretability, and usability is essential. In nutritional applications, this involves combining intake pattern analysis with requirement-based evaluation and predictive modeling to generate meaningful insights rather than isolated outputs [23], [24].

Therefore, the proposed Predictive Health Intervention Framework is formulated as a hybrid architecture that synthesizes concepts from machine learning, nutritional science, and decision-support systems. The framework integrates three key components: intake pattern analysis, deficiency risk prediction, and nutrient adequacy evaluation within a unified pipeline. This integration enables the system to produce outputs that are predictive, interpretable, and tailored to individual user profiles.

By combining predictive intelligence with domain-informed nutritional evaluation, the proposed framework addresses the limitations of existing systems and provides a practical solution for preventive healthcare. Rather than directly prescribing dietary plans, the framework supports informed decision-making by identifying nutrient deficiencies and presenting interpretable insights. This approach enhances usability while maintaining technical reliability, making the system suitable for real-world nutritional monitoring and guidance.

3.6.2 Layered Architecture and Operational Mechanism

The proposed Predictive Health Intervention Framework (PHIF) operates as a structured multi-layered architecture that transforms raw dietary intake data into meaningful and interpretable nutritional insights through sequential analytical stages. This layered design ensures that each stage contributes to improving prediction accuracy, consistency, and interpretability. Unlike conventional systems that rely on isolated processing steps, the proposed framework integrates multiple interconnected layers to progressively refine data before generating final outputs.

The first stage of the framework is the Intake Processing and Feature Generation Layer, where raw user input is transformed into structured nutritional data. In this stage, user-provided dietary intake, including food items and portion sizes, is mapped to corresponding nutrient values using the food nutrition dataset. These values are aggregated to compute total nutrient intake for key indicators such as energy, protein, calcium, and iron. Additionally, derived features such as nutrient adequacy ratios (intake-to-requirement) are computed. This layer ensures that unstructured intake data is converted into meaningful representations suitable for analytical processing.

The second stage is the Nutrient Comparison Layer, which acts as a domain-informed evaluation component within the framework. In this stage, aggregated nutrient intake values are compared against recommended nutrient requirements derived from age-based standards and health-condition-aware adjustments. This comparison identifies gaps between actual intake and required levels and provides a scientifically grounded basis for subsequent prediction.

The third stage is the Feature Representation Layer, where all processed features are structured into a consistent format for model input. This includes user demographic features (age, gender, health condition), nutrient intake values, requirement values, and adequacy ratios. The use of adequacy ratios is particularly important, as they normalize nutrient intake across different users and serve as the most informative indicators of deficiency risk.

Following this, the processed feature set is passed to the Machine Learning Prediction Layer, which forms the core analytical component of the framework. The system utilizes a Gradient Boosting classification model to predict deficiency risk levels categorized as LOW, MEDIUM, or HIGH. The model captures relationships between nutrient intake patterns and deficiency risk, enabling early identification of nutritional imbalances while maintaining interpretability.

The final stage of the framework is the Output Delivery Layer, where prediction results are presented to the user through the system interface. Outputs include deficiency risk levels, nutrient intake summaries, and visual insights displayed through a nutritional dashboard. The system emphasizes interpretability by presenting results in a clear and user-friendly manner, allowing users to understand their nutritional status and identify gaps in their diet.

Overall, the layered architecture of PHIF ensures that dietary intake data is transformed into reliable and interpretable insights through a sequence of structured processing stages. By integrating data-driven prediction with domain-informed nutritional evaluation, the framework provides a practical decision-support mechanism for identifying nutritional deficiencies and supporting preventive healthcare. This multi-layered design enhances prediction reliability while maintaining clarity and usability in real-world applications.

3.6.3 Technical Advantages and Research Contribution

The introduction of the Predictive Health Intervention Framework (PHIF) provides several significant technical advantages that distinguish the proposed system from existing nutritional guidance and dietary assessment approaches.

First, PHIF enhances predictive effectiveness by integrating machine learning-based deficiency risk prediction with domain-specific nutritional evaluation. Unlike traditional systems that rely solely on static guidelines or standalone predictive models, the proposed framework combines intake analysis with requirement-based comparison to ensure that predictions are both data-driven and context-aware. This hybrid approach improves the reliability and relevance of deficiency risk classification, addressing key limitations in existing nutritional systems.

Second, the framework improves robustness and consistency in nutritional assessment. Dietary intake data is inherently variable and influenced by user behavior, portion estimation, and food diversity. By incorporating structured preprocessing, nutrient aggregation, and requirement-based comparison, the system ensures that variations in raw input data are normalized before prediction. Additionally, the use of a Gradient Boosting model enhances robustness by effectively capturing complex patterns while maintaining generalization across diverse user profiles.

Third, PHIF demonstrates strong alignment with hybrid intelligent system principles, where multiple analytical components are integrated to improve system performance and interpretability. The combination of intake processing, nutrient comparison, adequacy ratio computation, and machine learning prediction enables the system to capture complex relationships between dietary patterns and nutritional outcomes. This layered integration enhances consistency and supports reliable operation in real-world scenarios characterized by variability and uncertainty [23].

Fourth, the framework enables decision-support-oriented processing by transforming predictive outputs into interpretable insights. Unlike many existing systems that provide only numerical outputs, PHIF presents deficiency risk levels (LOW, MEDIUM, HIGH) along with

structured nutritional summaries. This approach allows users to understand their nutritional status and identify gaps in their diet without requiring complex interpretation, thereby improving usability and practical applicability [24].

Fifth, PHIF significantly enhances personalization and contextual awareness. By incorporating age-based nutrient requirements and health-condition-aware adjustments, the system ensures that predictions are tailored to individual users. This level of personalization addresses a major limitation in existing systems, which often provide generalized outputs that do not reflect user-specific needs. As a result, the framework improves both the relevance and effectiveness of nutritional assessment.

Sixth, the framework improves interpretability and user trust by presenting outputs in a clear and structured manner. Instead of relying on complex numerical predictions, the system categorizes deficiency risk into intuitive levels and provides visual insights through a nutritional dashboard. This transparency in processing and output presentation enhances user confidence and supports adoption in real-world applications.

Overall, the Predictive Health Intervention Framework represents a hybrid intelligent architecture that bridges the gap between nutritional data analysis and practical decision support. By combining machine learning-based prediction, domain-informed evaluation, and user-centered design, the framework provides a comprehensive and interpretable solution for deficiency risk assessment. This integrated approach enhances prediction reliability while ensuring that outputs remain meaningful, usable, and aligned with real-world dietary monitoring needs.

3.7 Adaptive Nutritional Personalization

3.7.1 Concept and Motivation of Adaptive Personalization

While the proposed Predictive Health Intervention Framework (PHIF) enables accurate and context-aware nutritional analysis and deficiency risk prediction, it primarily operates on static model inference based on user input data. However, in real-world scenarios, dietary behavior is highly dynamic and influenced by factors such as lifestyle changes, food preferences, cultural habits, and accessibility. This variability highlights the need for systems that can adapt to individual user patterns over time.

To address this limitation, an Adaptive Nutritional Personalization Layer (ANPL) is proposed as a conceptual extension to the system. The objective of this layer is to enhance personalization by incorporating user-specific behavioral patterns, historical intake data, and interaction trends. Unlike the core predictive framework, which operates on structured input features, this layer is designed to support future improvements in system adaptability and user-centric analysis.

The ANPL is intended to analyze historical intake records and identify recurring patterns in user dietary behavior. By observing trends such as consistent nutrient deficiencies or repeated intake patterns, the system can refine its understanding of individual user profiles. This allows the system to move beyond static analysis and incorporate temporal insights into nutritional evaluation.

Furthermore, adaptive personalization aligns with established principles in intelligent decision-support systems, where user-specific data is utilized to improve relevance and usability of outputs [23], [24]. In the context of nutritional guidance, this enables the system to provide more context-aware insights that reflect not only current intake but also long-term behavioral tendencies.

Although the current implementation does not include real-time model updating or continuous learning mechanisms, the inclusion of ANPL highlights a potential direction for extending the system toward dynamic personalization. Future enhancements may incorporate feedback-

driven learning, preference modeling, or adaptive weighting of features to better align predictions with individual user behavior.

In addition, adaptive personalization has the potential to improve system usability and engagement. Systems that align outputs with user-specific patterns are more likely to be perceived as relevant and useful, encouraging sustained interaction. This is particularly important in health-related applications, where long-term user engagement plays a critical role in achieving meaningful outcomes.

Overall, the Adaptive Nutritional Personalization Layer transforms the system from a purely predictive model into a scalable framework capable of supporting future adaptive capabilities. While currently conceptual, this layer establishes a foundation for enhancing personalization, usability, and long-term effectiveness in real-world nutritional guidance systems.

3.7.2 Operational Mechanism and Learning Process

The Adaptive Nutritional Personalization Layer (ANPL) is designed as a conceptual behavior-driven refinement mechanism intended to enhance the relevance and effectiveness of system outputs based on user interaction patterns and historical intake behavior. While the current system primarily operates as a predictive decision-support model, this layer outlines a potential extension toward adaptive and user-centric personalization.

The proposed mechanism assumes that user dietary intake and interaction patterns can be recorded over time to identify behavioral trends. In such a setting, user responses to system outputs—such as repeated intake patterns or consistency in dietary behavior—can be analyzed to evaluate how closely system insights align with user habits. This behavioral information can then be used to refine future outputs, enabling the system to better reflect individual preferences and consumption tendencies.

To quantify this behavior, a conceptual adherence metric can be defined to represent the degree of alignment between recommended nutritional improvements and actual user intake patterns. This can be expressed as:

$$Adherence = ObservedAlignedIntake / TotalRelevantIntake$$

This metric provides an indication of how closely user behavior aligns with nutritionally desirable patterns. A higher adherence value suggests stronger alignment with nutritional goals, while a lower value indicates the presence of persistent intake gaps.

Based on this concept, a preference adjustment mechanism can be formulated to incorporate behavioral tendencies into future system outputs. A generalized formulation for such an adjustment is:

$$NewPreferenceWeight = (PreviousWeight \times retention_factor) + (Adherence \times learning_rate)$$

where:

- retention_factor represents how much past behavior is preserved
- learning_rate controls how strongly recent behavior influences the update
- Adherence reflects how closely user intake aligns with nutritional expectations

This formulation illustrates how a system could gradually adapt to user behavior while maintaining stability over time. By balancing historical patterns with recent observations, the mechanism avoids abrupt changes and ensures consistent personalization.

In addition, the proposed adaptive layer can incorporate a progressive learning strategy in which the influence of new behavioral data decreases as more interaction history is accumulated. This aligns with principles of learning stability and convergence, where early observations contribute more significantly to model adaptation, while later updates refine outputs incrementally [25].

A hierarchical personalization strategy can also be considered within this framework. When sufficient user-specific data is available, personalization can be performed at the individual level. In early stages, where limited data exists, the system can rely on generalized patterns derived from population-level data. This ensures that the system remains functional while gradually improving personalization as more data becomes available.

It is important to note that the current implementation of the system does not include real-time adaptive learning or feedback-driven model updates. The ANPL is therefore presented as a conceptual extension that highlights potential directions for enhancing personalization and adaptability in future versions of the system.

Overall, the proposed operational mechanism of ANPL provides a structured approach for incorporating behavioral insights into nutritional analysis. By integrating adherence estimation, preference adjustment, and hierarchical learning strategies, this layer establishes a foundation for future adaptive capabilities that can improve the relevance, usability, and long-term effectiveness of nutritional guidance systems.

3.7.3 Integration with PHIF and System Workflow

The Adaptive Nutritional Personalization Layer (ANPL) is conceptually integrated within the overall Predictive Health Intervention Framework (PHIF) as a post-analysis enhancement component. It is positioned after the core analytical stages of intake processing, nutrient comparison, and deficiency risk prediction. This placement ensures that personalization can be applied to refine system outputs while preserving the integrity and reliability of the underlying nutritional analysis.

Within the proposed workflow, ANPL is designed to interact with the outputs generated by the machine learning prediction layer. The deficiency risk levels produced by the system can be further interpreted in the context of user-specific behavioral patterns and historical intake trends. This design allows personalization to enhance the relevance of outputs without altering the core prediction logic. In this sense, the personalization layer focuses on improving how insights are presented and interpreted rather than modifying the fundamental nutritional requirements.

From a system perspective, ANPL introduces a conceptual feedback-driven mechanism that leverages historical intake data and user interaction patterns. These interactions, such as recurring dietary habits or consistent intake trends, can be stored within the system database and used to support future personalization. Unlike real-time adaptive learning systems, the

current implementation does not include continuous model updates; however, the framework establishes a foundation for incorporating such capabilities in future extensions.

The integration of this conceptual feedback mechanism allows the system to evolve toward a more user-centric design. As more user data becomes available, personalization strategies can be refined to better reflect real-world dietary behavior. This approach aligns with modern intelligent system architectures, where user interaction data is utilized to enhance long-term system relevance and usability [23].

Furthermore, the integration of ANPL within PHIF is designed to ensure compatibility with other system components, including data processing, feature engineering, and prediction modules. Personalization is applied in a controlled manner, ensuring that outputs remain consistent with established nutritional requirements and domain constraints. This separation between prediction and personalization enhances system transparency and prevents unintended distortion of core analytical results.

It is important to note that the current system operates as a predictive decision-support platform, and the adaptive personalization layer is proposed as a conceptual extension rather than a fully implemented module. Future enhancements may incorporate feedback-driven learning, preference modeling, and adaptive feature weighting to enable dynamic personalization.

Overall, the integration of the Adaptive Nutritional Personalization Layer within the system workflow establishes a scalable framework capable of supporting future adaptive capabilities. By combining predictive modeling with user-centered personalization concepts, the proposed architecture enhances the potential for improved usability, relevance, and real-world applicability while maintaining reliability and interpretability.

3.7.4 Technical Advantages and Contribution

The inclusion of the Adaptive Nutritional Personalization Layer (ANPL), as a conceptual extension, provides several potential advantages that can enhance the overall effectiveness of the proposed Nutritional Guidance – Predictive Health Intervention System.

First, ANPL introduces the capability for enhanced personalization by incorporating user-specific behavioral patterns into nutritional analysis. While the current system provides predictions based on intake and requirement comparison, the proposed personalization layer can further refine outputs by considering historical intake trends and user-specific dietary behavior. This approach allows the system to move beyond generalized analysis toward more context-aware insights.

Second, the framework supports improved long-term effectiveness through the potential use of historical intake data. By analyzing accumulated user data over time, the system can identify recurring nutritional gaps and patterns. Although continuous learning is not implemented in the current system, this conceptual mechanism highlights how future enhancements could improve alignment between system outputs and real-world dietary behavior.

Third, ANPL contributes to system adaptability by providing a foundation for handling diverse user profiles and dietary patterns. Nutritional behavior varies significantly across individuals due to cultural, social, and lifestyle factors. The proposed adaptive layer allows the system to evolve toward accommodating such variability by incorporating user-specific trends in future implementations.

Fourth, the personalization layer has the potential to improve user engagement and usability. Systems that reflect user-specific patterns are more likely to be perceived as relevant and practical. By aligning outputs with user behavior, the framework can enhance user satisfaction and encourage sustained interaction, which is essential in health-related applications.

Fifth, the integration of ANPL within the Predictive Health Intervention Framework demonstrates a hybrid intelligent architecture that combines intake analysis, requirement-based evaluation, and predictive modeling with a proposed personalization extension. This

layered design represents an advancement over traditional systems that focus solely on static analysis or isolated predictive components.

Overall, the inclusion of adaptive personalization transforms the framework into a scalable architecture capable of supporting future enhancements in user-centric analysis. While the current implementation operates as a predictive decision-support system, the ANPL establishes a foundation for incorporating behavioral adaptation in subsequent versions. This approach enhances the long-term potential of the system by improving usability, relevance, and real-world applicability in nutritional guidance.

3.8 Smart Nutritional Recommendation Priority

3.8.1 Concept and Motivation of SNRP

While accurate deficiency prediction is a key capability of the proposed system, translating these predictions into practical user actions remains a challenge. In real-world scenarios, users may find it difficult to interpret deficiency information and determine how to prioritize improvements in their dietary intake. Simply identifying nutrient gaps does not necessarily provide clear guidance on which aspects of nutrition should be addressed first.

To address this limitation, the Smart Nutritional Recommendation Priority (SNRP) module is proposed as a conceptual decision-support extension that enhances the interpretability of deficiency predictions. The objective of SNRP is to provide a structured approach for prioritizing nutritional improvements based on the severity of identified deficiencies. Instead of treating all nutrient gaps equally, this approach enables users to focus on the most critical deficiencies first.

The proposed prioritization mechanism is based on the relative severity of nutrient deficiencies derived from model predictions and adequacy ratios. Nutrients with lower adequacy levels indicate higher deficiency risk and can therefore be considered higher priority for improvement. This concept aligns with established principles in intelligent decision-support systems, where analytical outputs are enhanced through structured ranking mechanisms to improve decision-making efficiency [24].

Furthermore, integrating prioritization into predictive nutritional systems reflects broader trends in data-driven decision support, where model outputs are transformed into actionable insights. In this context, prioritization does not involve generating new recommendations but rather organizing existing insights in a way that supports clearer user understanding and decision-making [25].

In addition, prioritization has the potential to improve usability by simplifying how users interpret system outputs. When nutritional insights are structured based on importance, users

can more easily identify which areas require immediate attention. This reduces cognitive complexity and enhances the practical value of the system in real-world scenarios.

It is important to note that the current implementation of the system focuses on deficiency risk prediction and nutritional analysis. The SNRP module is therefore presented as a conceptual extension that outlines how prioritization mechanisms can be incorporated in future enhancements to improve decision support and user guidance.

Overall, the Smart Nutritional Recommendation Priority module extends the proposed framework by introducing a structured approach to interpreting deficiency predictions. By prioritizing nutritional insights based on severity, the system can evolve toward providing clearer, more actionable guidance while maintaining interpretability and alignment with real-world user needs.

3.8.2 Priority Scoring Mechanism

The Smart Nutritional Recommendation Priority (SNRP) module is designed as a conceptual mechanism to organize deficiency-related insights based on their importance. The main idea is to help users understand which nutritional gaps should be focused on first, rather than treating all deficiencies equally.

Instead of treating all nutrient deficiencies in the same way, this approach introduces a simple prioritization strategy based on two key factors:

- The deficiency risk level predicted by the machine learning model (LOW, MEDIUM, HIGH)
- Nutrient gap, which represents how far current intake is from the required level

To keep the system easy to understand, a simple scoring rule is used to represent priority levels:

- HIGH risk - 1.0 (high priority)
- MEDIUM risk - 0.7 (medium priority)
- LOW risk - 0.4 (low priority)

Nutrient Gap Calculation:

To further support prioritization, the system also considers how large the deficiency is. This is calculated using a simple ratio:

$$Gap(x) = (Required(x) - Intake(x)) / Required(x)$$

This value shows how far the user is from meeting the required intake:

A higher value - larger deficiency

A lower value - closer to requirement

In this approach:

- The risk level gives a general priority category (LOW / MEDIUM / HIGH)
- The gap value helps understand the severity within that category

For example, two nutrients may both be classified as HIGH risk, but the one with a larger gap can be considered more important.

It is important to note that this prioritization mechanism is presented as a conceptual decision-support extension. The current system focuses on deficiency risk prediction and nutritional analysis. The SNRP logic demonstrates how prediction outputs can be structured in a more meaningful way for users in future enhancements.

Overall, the SNRP mechanism provides a simple and interpretable way to organize nutritional insights. By combining predicted risk levels with intake gap analysis, it allows users to better understand which deficiencies require more attention, while keeping the system easy to use and explain.

3.8.3 Integration with System Workflow

The Smart Nutritional Recommendation Priority (SNRP) module is conceptually positioned as the final decision-support layer within the overall Predictive Health Intervention Framework (PHIF). It is designed to operate after the core analytical stages of intake processing, nutrient comparison, and deficiency risk prediction. This placement ensures that prioritization is based on structured and reliable outputs generated by the system.

Within the proposed workflow, SNRP utilizes the predicted deficiency risk levels and nutrient adequacy information produced by the machine learning model. Rather than generating new outputs, the module focuses on organizing existing insights based on their relative importance. This ensures that prioritization reflects the severity of nutritional deficiencies while maintaining consistency with the underlying analytical results.

The prioritization mechanism can be used to structure system outputs in a more interpretable way within the user interface. For example, deficiency risks associated with higher severity can be highlighted more prominently, while lower-risk insights can be presented as secondary information. This structured presentation simplifies decision-making and helps users quickly identify which nutritional gaps require more attention.

In addition, SNRP can be integrated with visualization components such as the nutritional dashboard to improve clarity and usability. By presenting deficiency levels alongside relative importance, users can better understand their nutritional status and interpret system outputs more effectively. This approach enhances transparency by linking analytical results with structured insights.

From a system perspective, SNRP represents a conceptual extension that completes the transformation of analytical outputs into decision-support information. While the current system focuses on intake analysis and deficiency risk prediction, the inclusion of prioritization highlights how these outputs can be organized to improve real-world usability. This aligns with modern intelligent system design principles, where predictive analytics is complemented by decision-support mechanisms to enhance interpretability and user experience [23].

It is important to note that the SNRP module is not fully implemented in the current system and is presented as a proposed enhancement for future development. Its integration demonstrates how prioritization mechanisms can be incorporated to improve the practical applicability of predictive nutritional systems.

Overall, the conceptual integration of SNRP within the system workflow establishes a scalable architecture that supports future decision-support capabilities. By structuring deficiency-related insights based on importance, the framework enhances usability, clarity, and potential real-world impact while maintaining consistency with the core predictive model.

3.9 Tools and Technologies

The development and implementation of the proposed Nutritional Guidance – Predictive Health Intervention System required the integration of multiple tools and technologies spanning machine learning, backend development, database management, and frontend user interaction. These technologies were selected to ensure efficiency, scalability, and seamless integration within a unified full-stack architecture.

The machine learning component of the system was developed using Python, which provides a comprehensive ecosystem for data analysis, model development, and scientific computing. Libraries such as Pandas and NumPy were utilized for data preprocessing, transformation, and aggregation of nutritional datasets. These libraries enabled efficient handling of structured data, including food composition data, nutrient requirements, and user intake logs. The predictive model was implemented using a Gradient Boosting classification algorithm from the Scikit-learn library, which is well-suited for structured tabular data. Gradient Boosting was selected due to its ability to capture complex nonlinear relationships, improve predictive accuracy through sequential learning, and maintain strong generalization performance in classification tasks [26].

For backend development, the system utilizes the Flask framework, a lightweight and flexible web framework for building RESTful APIs. Flask serves as the core communication layer between the frontend interface and the machine learning model, enabling real-time data processing and prediction. The backend is responsible for handling user requests, processing dietary intake data, executing the predictive pipeline (including intake processing, nutrient comparison, feature engineering, and deficiency risk prediction), and returning structured results to the user interface. The modular design of Flask supports easy integration of system components while maintaining scalability.

The system employs MongoDB as its primary database for storing user profiles, dietary intake logs, and prediction results. MongoDB's NoSQL architecture provides flexibility in schema design, which is particularly beneficial for handling dynamic and user-generated data. The integration of MongoDB with Flask using libraries such as Flask-PyMongo enables efficient

data storage and retrieval. This allows the system to maintain historical intake data and support analytical processing over time.

On the frontend, the system is developed using React, a modern JavaScript library for building dynamic and interactive user interfaces. React enables the creation of reusable components that allow users to input dietary data, visualize nutritional insights, and view deficiency risk outputs. The frontend communicates with the backend through RESTful API calls, ensuring a responsive user experience. Additional tools such as Axios are used for handling HTTP requests, while standard web technologies including HTML and CSS are utilized for layout design and styling. The interface is designed to present nutritional data in a clear and user-friendly manner.

The development process was supported by tools such as GitHub for version control and collaboration, ensuring proper tracking of code changes. Python virtual environments were used to manage dependencies and maintain consistency across system modules, improving reproducibility and stability during development.

Overall, these tools and technologies enable the implementation of a full-stack intelligent system that integrates machine learning, data processing, and user interaction within a cohesive platform. The use of widely adopted and scalable technologies ensures that the system is efficient, reliable, and suitable for extension in real-world nutritional guidance applications.

3.10 Commercialization Aspect of the Product

The proposed Nutritional Guidance – Predictive Health Intervention System demonstrates strong commercialization potential due to its ability to address real-world challenges related to nutritional imbalance, preventive healthcare, and data-driven dietary monitoring. With the increasing global focus on health awareness, lifestyle diseases, and digital healthcare solutions, there is a growing demand for intelligent systems that provide personalized and actionable nutritional insights [1]. Unlike conventional dietary tracking applications, the proposed system integrates predictive analytics with structured nutritional analysis, offering a practical solution for both individual users and healthcare-related industries.

At the individual level, the system can be commercialized as a web-based or mobile application that enables users to monitor their dietary intake, assess nutritional status, and understand potential deficiency risks. By analyzing intake patterns and predicting deficiency levels, the system provides proactive health insights rather than relying on static tracking. Features such as nutritional dashboards, intake summaries, and risk-level visualization enhance usability and encourage users to make informed dietary decisions.

Beyond individual usage, the system has strong applicability in healthcare and wellness sectors, including hospitals, fitness centers, diet clinics, and telemedicine platforms. Healthcare professionals can utilize the system as a decision-support tool to monitor patient dietary behavior and identify potential nutritional deficiencies. By providing data-driven insights, the system supports more efficient nutritional assessment compared to manual methods. This aligns with the growing trend of integrating artificial intelligence into healthcare systems to enhance preventive care and early intervention.

In addition, the system can be extended to corporate wellness programs and institutional environments such as schools and workplaces. Organizations can leverage the system to promote healthier eating habits by analyzing dietary patterns and identifying common nutritional gaps among groups. This can contribute to improved well-being, reduced health risks, and increased productivity. Aggregated insights can also support the design of targeted health initiatives.

Furthermore, the system has potential for integration into broader digital health ecosystems, including wearable devices and fitness tracking platforms. By combining dietary intake analysis with activity data, the system can evolve toward providing more comprehensive health insights. Such integrations represent a future extension that aligns with emerging trends in AI-driven health monitoring and personalized wellness systems.

From a business model perspective, the system supports multiple monetization strategies. These include subscription-based services for individual users, enterprise solutions for healthcare providers and wellness organizations, and API-based integration for third-party platforms such as fitness applications and telemedicine systems. The modular architecture of the system allows flexible deployment and scalability across different use cases.

It is important to note that advanced features such as adaptive personalization and prioritization mechanisms are proposed as future enhancements. While the current system focuses on deficiency risk prediction and nutritional analysis, these extensions can further improve user engagement and decision support in subsequent versions.

Overall, the proposed system offers a combination of predictive intelligence, structured nutritional evaluation, and user-friendly insights, positioning it as a viable solution in the digital health and nutrition technology market. By addressing both individual and institutional needs, the system demonstrates strong potential for commercialization and real-world impact in supporting preventive healthcare and healthier lifestyles.

3.10.1 Future Scope

While the proposed Nutritional Guidance – Predictive Health Intervention System demonstrates strong performance and practical applicability, several opportunities exist for further enhancement and expansion.

One key direction for future work is the integration of real-time data from wearable devices and health monitoring systems. By incorporating data such as physical activity levels, calorie expenditure, and biometric indicators, the system can provide more dynamic and context-aware nutritional insights, improving overall accuracy and personalization.

Another potential enhancement is the incorporation of computer vision techniques for automated food recognition. By utilizing image-based models, users could capture images of their meals, and the system would automatically identify food items and estimate nutritional values. This would significantly reduce reliance on manual input and improve user convenience, making the system more accessible and user-friendly.

The personalization mechanism can also be extended through the use of advanced adaptive learning techniques, such as reinforcement learning or online learning models. These approaches would enable the system to better utilize historical intake data and user behavior patterns to refine outputs over time. This would allow the system to evolve into a more intelligent assistant capable of understanding complex dietary patterns.

Additionally, future work can explore deeper integration with healthcare systems and medical records. By linking nutritional analysis with clinical data, the system can provide more precise and context-aware dietary insights tailored to specific medical conditions. This would enhance its applicability in clinical settings and support healthcare professionals in nutritional assessment.

From a scalability perspective, the system can be expanded to support multi-user environments and large-scale deployments, particularly in healthcare institutions, fitness platforms, and wellness programs. Cloud-based infrastructure and distributed computing techniques can be

utilized to improve system performance, reliability, and accessibility across different platforms.

Finally, future research can focus on improving model interpretability and explainability. Providing clear explanations for deficiency predictions and analytical outputs will enhance user trust and adoption. This is particularly important in health-related systems, where users need to understand the reasoning behind AI-generated insights to make informed decisions [25].

3.11 Testing and Implementation

The proposed system was systematically tested to evaluate its accuracy, reliability, and usability. The testing process included multiple stages such as model evaluation, functional testing, API validation, and scenario-based testing to ensure consistent performance across different use cases.

3.11.1 Model Training and Evaluation

The dataset used for deficiency prediction was divided into training and testing subsets using an 80:20 split to ensure reliable performance evaluation. The Gradient Boosting classification model was trained on the training dataset to learn relationships between nutrient intake features and deficiency risk levels. These features included energy, protein, calcium, and iron intake values, along with derived nutrient adequacy ratios.

The trained model was evaluated on unseen test data to assess its generalization capability. Model performance was measured using classification metrics such as accuracy, precision, recall, weighted F1-score, and cross-validation. The best-performing model was Gradient Boosting, achieving an accuracy of 84.38% and a weighted F1-score of 0.8446. The 5-fold cross-validation produced an average F1-score of approximately 0.8286 ± 0.0280, indicating reliable predictive performance.

These results demonstrate that the model effectively captures relationships between dietary intake patterns and nutritional deficiency risk, making it suitable for predictive health analysis and decision-support applications.

3.11.2 Functional Testing

Functional testing was conducted to verify that all system components operate as expected. The intake processing module was tested to ensure accurate aggregation of nutrient values from user input. The nutrient comparison layer was validated to confirm correct calculation of intake versus requirement gaps.

The machine learning prediction module was tested to ensure accurate classification of deficiency risk levels based on processed input features. Additionally, the system outputs were validated to ensure that deficiency levels and nutritional summaries were correctly generated and displayed in the user interface.

3.11.3 API and Integration Testing

All backend API endpoints were tested using both the frontend interface and API testing tools. The system was validated to ensure correct responses for valid inputs, while appropriate error handling mechanisms were triggered for invalid or incomplete data.

Integration testing was performed to ensure seamless communication between the frontend, backend, machine learning model, and database. This confirmed that the system operates as a unified platform with efficient and consistent data flow across all components.

3.11.4 Scenario-Based Testing

To evaluate real-world applicability, scenario-based testing was conducted. For example, a user inputs daily dietary intake, and the system processes the data to calculate nutrient consumption, compare it with required levels, and predict deficiency risk categories.

Multiple test scenarios were evaluated with varying intake patterns to ensure that the system produces consistent and interpretable outputs. These scenarios confirm that the system can effectively support real-world nutritional monitoring and decision-making.

3.11.5 Edge Case Testing

Edge cases were tested to ensure system robustness under unusual conditions. These included incomplete dietary inputs, missing nutrient values, and extreme intake patterns. The system successfully handled these scenarios by applying default values, validation checks, and fallback mechanisms, ensuring stable and reliable performance.

3.11.6 User Interface Validation

The frontend interface was tested to ensure usability, clarity, and responsiveness. Users were able to input dietary data, view deficiency risk levels, and interpret nutritional insights without confusion. Visual elements such as dashboards, charts, and summaries were validated to ensure effective communication of system outputs.

4. RESULTS AND DISCUSSION

4.1 Results

The performance of the proposed Nutritional Guidance – Predictive Health Intervention System was evaluated through dietary intake trend analysis, nutrient adequacy assessment, and machine learning-based deficiency prediction. The results demonstrate the effectiveness of combining data-driven modeling with domain-specific nutritional evaluation to support accurate deficiency risk analysis.

As illustrated in Figure 4.1, the average weekly intake of key nutrients such as energy, protein, calcium, and iron shows noticeable variability across different weeks. A significant increase in energy and calcium intake is observed during mid-January, indicating irregular dietary patterns. While some weeks reflect relatively adequate nutrient intake, others show reduced consumption levels, particularly for protein and iron. These fluctuations highlight inconsistencies in dietary habits and directly influence the nutrient adequacy ratios used as inputs for the machine learning model. Periods of lower intake correspond to higher predicted deficiency risks, demonstrating the system's ability to capture meaningful relationships between dietary patterns and nutritional status

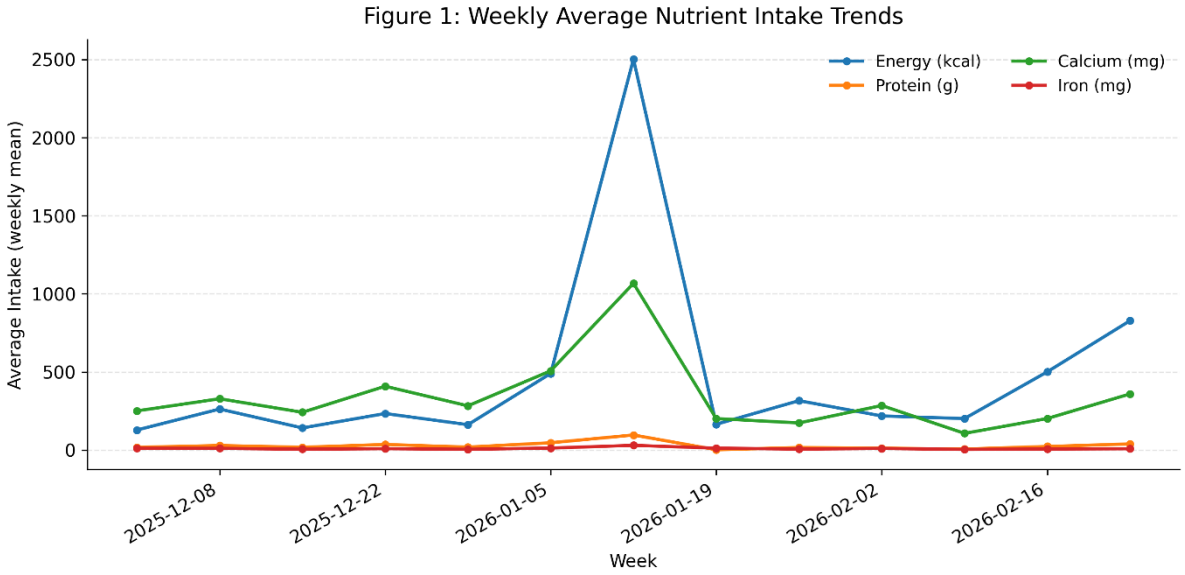

Figure 4.1 – Weekly Nutrition Intake Trends

The comparison between average intake and recommended nutritional requirements is presented in Figure 4.2, where the results indicate that the average intake of energy (493 kcal) and calcium (355 mg) is considerably lower than the recommended levels (1600 kcal and 1000 mg, respectively), reflecting a significant nutritional gap and a potential risk of deficiency. In contrast, protein intake slightly exceeds the recommended value, while iron intake remains approximately at the required level. These differences between intake and recommended values are important for calculating nutrient adequacy ratios, which are used as key input features for the machine learning model. Lower intake relative to requirements results in reduced ratio values, contributing to higher predicted deficiency risk levels, thereby demonstrating the importance of nutrient-specific evaluation for accurate nutritional assessment.

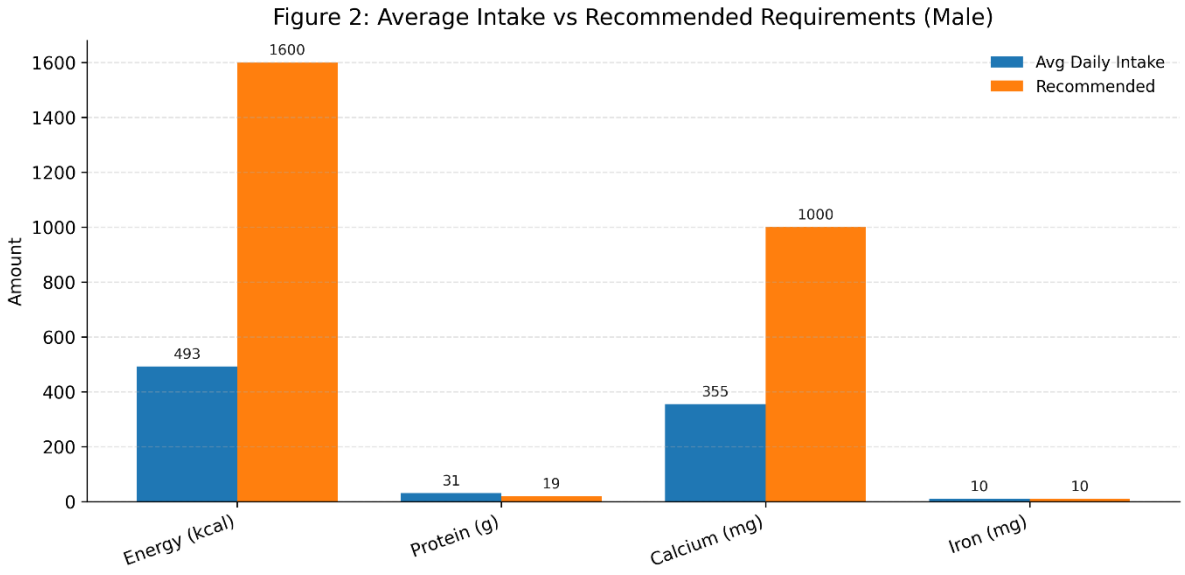

Figure 4.2 – Nutrition Intake vs Recommended Values

The nutrient adequacy ratio trends shown in Figure 4.3 provide further insight into the relationship between nutrient intake and recommended requirements over time. The dashed reference line at 1.0 represents the threshold for adequate intake, where values below this level indicate insufficient consumption. The majority of observed values fall below this threshold, particularly for calcium and iron, highlighting consistent under-consumption of these nutrients. Although occasional spikes above the threshold are observed, these are irregular and do not indicate sustained adequate intake. These fluctuations directly influence the nutrient

adequacy ratios used as key input features for the machine learning model, where lower ratio values contribute to higher predicted deficiency risk levels. This analysis demonstrates that nutritional imbalance is both persistent and variable, reinforcing the importance of continuous monitoring and data-driven deficiency assessment.

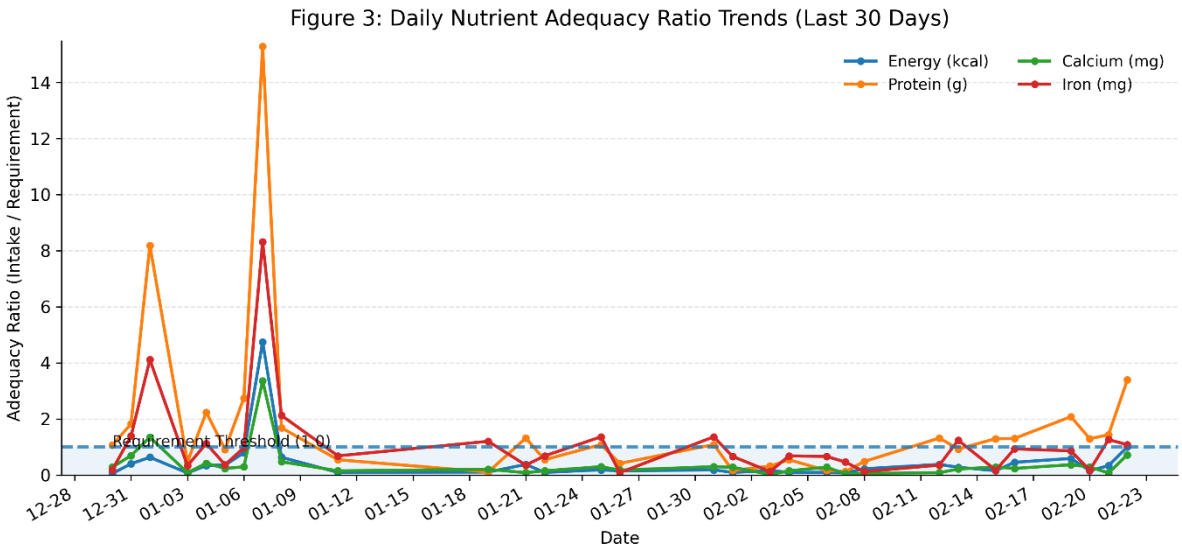

Figure 4.3 – Daily Nutrition Adequacy Ratio Trends

The distribution of predicted deficiency risk levels is illustrated in Figure 4.4**.** The results indicate that a higher proportion of observations fall into the HIGH risk category (30 cases), followed by LOW (13 cases) and MEDIUM (9 cases). This distribution suggests that, within the analyzed dataset, nutrient intake is frequently below recommended levels, leading to elevated deficiency risk classifications. The predominance of HIGH risk cases reflects the impact of consistently low nutrient adequacy ratios observed in earlier analyses. These results demonstrate the model's ability to effectively classify deficiency risk based on intake patterns and highlight the importance of continuous monitoring for identifying potential nutritional imbalances.

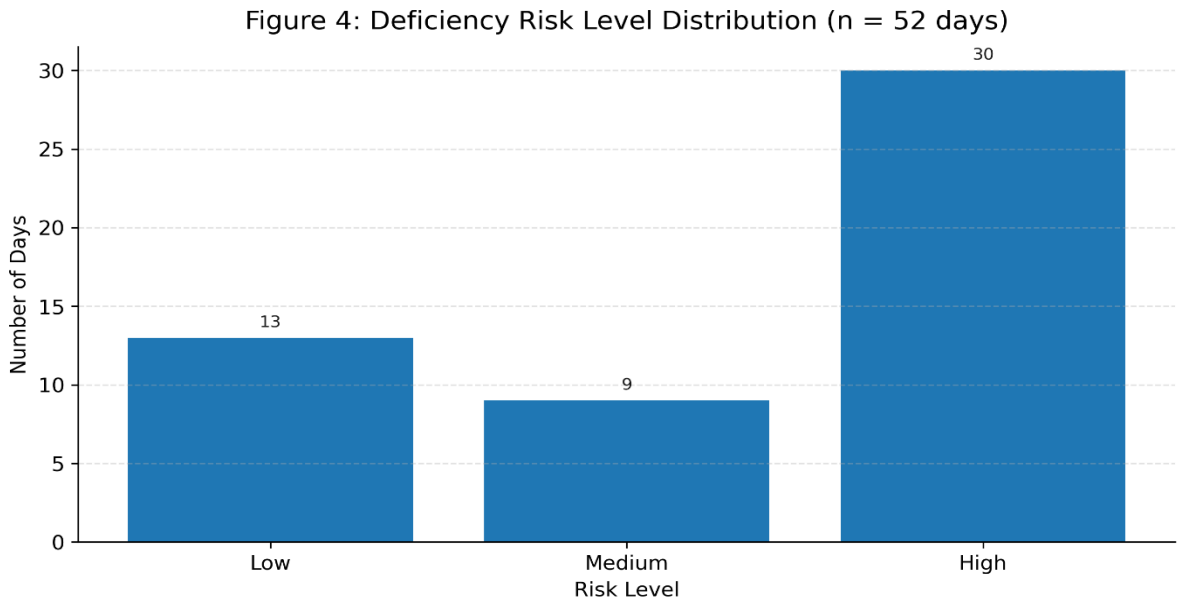

Figure 4.4 – Deficiency Risk Distribution

The predictive performance of the machine learning model is evaluated using the confusion matrix presented in Figure 4.5, which shows strong classification performance across all three risk categories, with most predictions concentrated along the diagonal. For the LOW risk class, 77 instances are correctly identified, with only a few misclassified as MEDIUM. For the MEDIUM class, 269 instances are correctly classified, with some misclassifications into LOW (13) and HIGH (24), indicating moderate overlap between neighboring categories. Similarly, for the HIGH risk class, 218 instances are correctly identified, with a limited number misclassified as MEDIUM (33). These misclassifications mainly occur between adjacent classes (LOW–MEDIUM and MEDIUM–HIGH), which is expected due to gradual transitions in nutrient adequacy levels, while no direct misclassification occurs between LOW and HIGH classes. Overall, the confusion matrix demonstrates that the model maintains balanced and reliable performance in distinguishing deficiency risk levels, confirming its suitability for nutritional risk prediction.

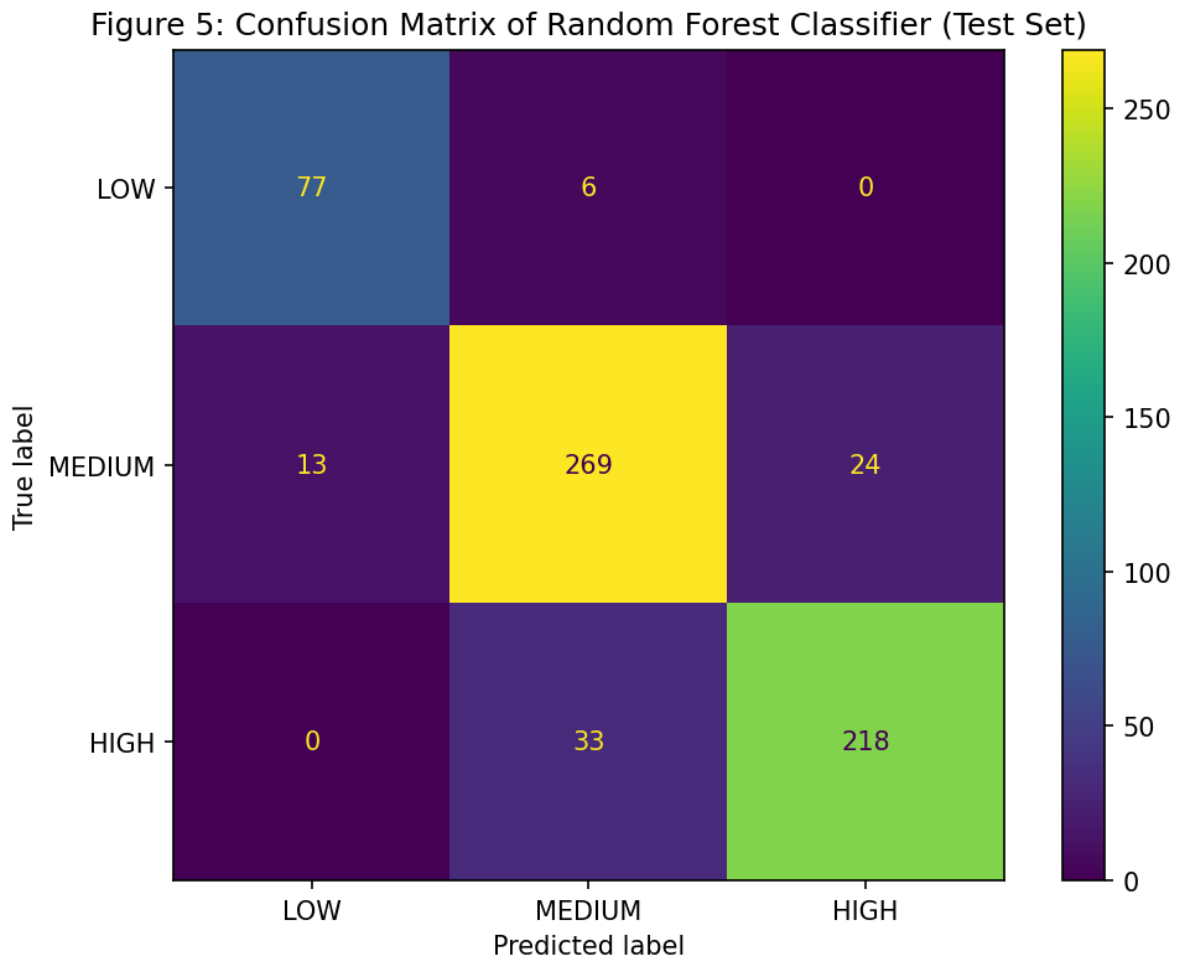

Figure 4.5 – Confusion Matrix

Additional insights into model behavior are provided by the feature importance analysis shown in Figure 4.6. The results indicate that calcium intake is the most influential feature in predicting deficiency risk, followed by protein and iron intake, while energy intake also contributes to the model's predictions. In contrast, features such as age and health condition have comparatively lower influence, suggesting that the model relies more heavily on direct nutrient intake patterns than demographic factors. This aligns with the feature engineering approach, where nutrient values and adequacy-related inputs form the core predictors of deficiency risk. The higher importance assigned to calcium reflects its strong impact on adequacy ratios and classification outcomes, highlighting the role of nutrient-specific intake in driving the model's predictive behavior.

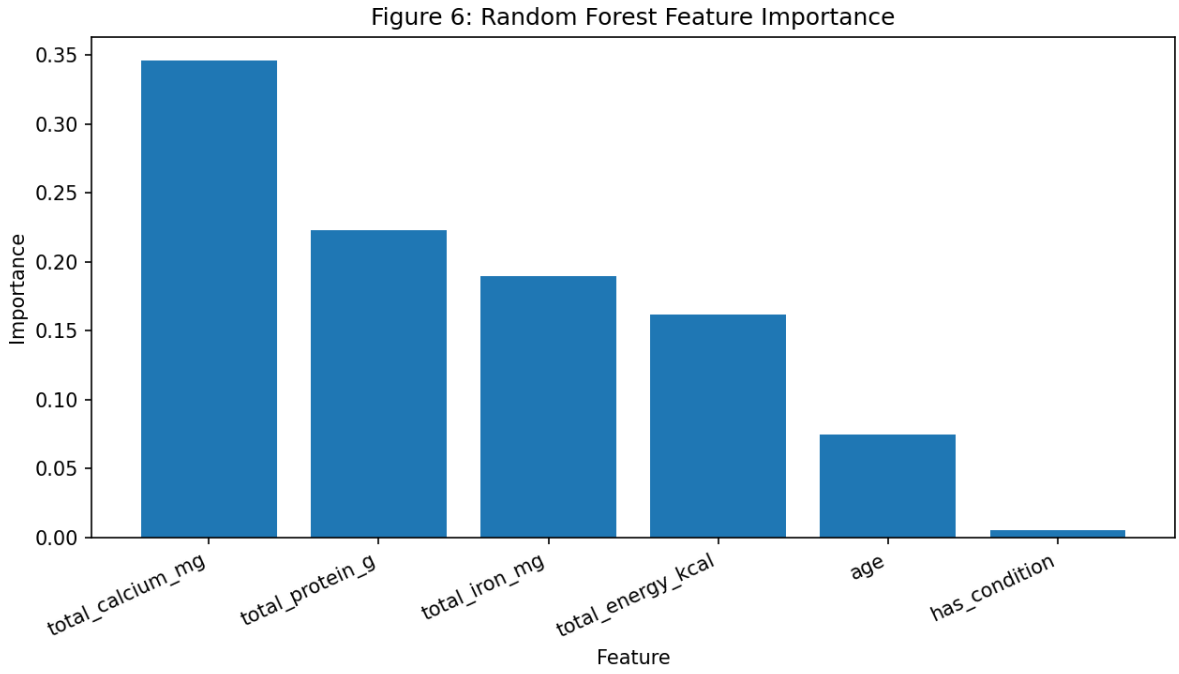

Figure 4.6 – Feature Importance analysis

Overall, the results demonstrate that the proposed system effectively integrates dietary intake analysis, feature-based evaluation, and machine learning to identify potential nutritional deficiencies. The combination of statistical analysis and predictive modeling enables the system to capture variations in dietary patterns and translate them into meaningful deficiency risk classifications. The results further indicate that the model provides consistent and interpretable outputs based on nutrient intake and adequacy ratios. These findings suggest that the system is reliable for nutritional assessment and can support data-driven analysis of dietary behavior in practical scenarios.

4.2 Research Findings

The experimental evaluation of the proposed Nutritional Guidance – Predictive Health Intervention System led to several key findings that validate both the machine learning model and the overall system design. These findings highlight the importance of integrating dietary intake patterns, nutrient-based analysis, and data-driven modeling within a unified predictive framework.

A primary finding of this study is that nutrient intake imbalance is a major factor influencing deficiency risk levels. As illustrated in Fig. 4.1, weekly nutrient intake trends show noticeable fluctuations across key nutrients such as energy, protein, calcium, and iron. This variability indicates inconsistent dietary habits, leading to irregular nutrient consumption. Such patterns reinforce the importance of continuous monitoring, as static dietary assessments are insufficient to capture dynamic intake behavior.

A second important finding is that nutrient deficiencies vary across different nutrients. The comparison between average intake and recommended requirements in Fig. 4.2 reveals significant gaps in nutrients such as energy and calcium, while protein intake remains relatively adequate. This demonstrates that nutritional assessment must be performed at an individual nutrient level rather than relying on overall intake measures, validating the system's nutrient-specific analysis approach.

Another key finding relates to the role of nutrient adequacy ratios in identifying deficiency trends. As shown in Fig. 4.3, most intake-to-requirement ratios remain below the threshold value of 1.0, particularly for calcium and iron, indicating consistent under-consumption. Although occasional increases above the threshold are observed, these are not sustained over time. This highlights the importance of temporal analysis, where trends provide more meaningful insights than isolated data points.

The distribution of deficiency risk levels (Fig. 4.4) shows that a larger proportion of observations fall into the HIGH risk category compared to LOW and MEDIUM levels. This suggests that, within the analyzed dataset, nutrient intake is frequently below recommended

levels. This finding emphasizes the importance of predictive systems that can identify potential deficiencies based on intake patterns.

The evaluation of the machine learning model demonstrates that the Gradient Boosting classifier is effective in predicting deficiency risk levels. The confusion matrix in Fig. 4.5 shows strong classification performance, with most predictions aligned along the diagonal. Misclassifications occur primarily between adjacent classes (e.g., MEDIUM and HIGH), which is expected due to gradual transitions in nutrient adequacy levels. This indicates that the model successfully captures relationships between intake patterns and deficiency risk.

Another important finding is the influence of specific nutrients on prediction outcomes. The feature importance analysis in Fig. 4.6 indicates that calcium intake is the most influential feature, followed by protein and iron, while energy intake also contributes to the model's predictions. In contrast, demographic features such as age and health condition have comparatively lower influence. This confirms that deficiency prediction is primarily driven by nutrient intake patterns, validating the feature engineering approach used in this study.

Overall, these findings confirm that accurate nutritional deficiency prediction requires a multi-dimensional approach combining intake analysis, temporal trend evaluation, and machine learning techniques. The proposed system effectively integrates these components, resulting in a reliable and interpretable framework for predictive nutritional assessment.

4.3 Discussions

The results of this study demonstrate that the proposed Nutritional Guidance – Predictive Health Intervention System effectively addresses key limitations of traditional dietary assessment approaches by integrating predictive modeling with nutrient-based evaluation. Unlike conventional systems that rely on static dietary guidelines or simple calorie tracking, the proposed framework combines intake analysis with machine learning to provide data-driven deficiency risk assessment.

One of the most significant observations from the results is the strong influence of dietary intake patterns on nutritional deficiency risk. The variability observed in weekly nutrient intake trends confirms that users do not maintain consistent dietary habits, leading to irregular nutrient consumption. This finding aligns with established nutritional principles, where sustained imbalance in nutrient intake contributes to deficiency-related health risks. The incorporation of temporal intake analysis therefore represents an important design choice, ensuring that predictions reflect real-world dietary behavior rather than isolated observations.

The analysis also highlights the importance of nutrient-specific evaluation in identifying deficiencies. Differences observed between nutrients such as calcium, energy, protein, and iron demonstrate that deficiency risk is not uniform across all nutrients. Certain nutrients, particularly calcium, consistently show lower intake levels, making them more influential in determining overall nutritional status. This supports the system's design, which focuses on individual nutrient analysis rather than aggregated dietary measures, thereby improving the precision and relevance of predictions.

A key contribution of this work is the integration of machine learning with domain-specific nutritional knowledge. The Gradient Boosting model effectively captures nonlinear relationships between nutrient intake patterns and deficiency risk levels, while the incorporation of requirement-based comparison ensures that predictions are aligned with established nutritional standards. This combination improves both predictive accuracy and interpretability, which are essential in health-related applications.

Despite these strengths, certain limitations must be acknowledged. The system relies on user-reported dietary intake, which may introduce inaccuracies due to estimation errors or incomplete data entry. Additionally, the dataset used for training, although structured, may not fully capture variations in dietary habits across different populations and contexts. These factors may affect the generalizability of the model.

These limitations provide opportunities for future improvement. The integration of wearable devices and automated food tracking technologies can enhance data accuracy and reduce reliance on manual input. Similarly, incorporating image-based food recognition techniques can improve usability. Advanced features such as adaptive personalization and prioritization mechanisms can also be explored in future work to enhance system functionality.

Overall, the findings confirm that the proposed system provides an effective framework for predictive nutritional assessment. By combining intake analysis, nutrient evaluation, and machine learning within a unified architecture, the system offers a reliable and interpretable approach for analyzing dietary patterns and identifying potential nutritional deficiencies

5. CONCLUSION

This research presented the design and development of a Nutritional Guidance – Predictive Health Intervention System that integrates machine learning with nutrient-based evaluation to address challenges related to nutritional imbalance and preventive healthcare. The study was motivated by the limitations of existing dietary assessment approaches, which often rely on static guidelines or lack context-aware analysis. To address these challenges, the proposed system combines dietary intake analysis with data-driven modeling to provide structured deficiency risk assessment.

At the core of the system is a Gradient Boosting classification model, which demonstrated reliable predictive performance in identifying nutritional deficiency risk levels. The model achieved an accuracy of approximately 84.38%**,** with a weighted F1-score of 0.8446 and a 5-fold cross-validation F1-score of approximately 0.8286 ± 0.0280, indicating consistent generalization across different data subsets. These results confirm the model's ability to capture relationships between dietary intake patterns and nutritional deficiencies. Compared to traditional rule-based systems, this approach provides more flexible and data-driven insights for deficiency risk classification.

A key contribution of this research is the development of a structured predictive framework that integrates intake processing, nutrient comparison, and machine learning within a unified architecture. By combining user intake data with recommended nutritional requirements, the system ensures that predictions are both data-driven and aligned with established dietary standards. This design improves the interpretability and relevance of deficiency risk predictions.

The results of this study demonstrate that nutritional deficiency risk is influenced by multiple factors, including intake variability and nutrient-specific imbalances. The integration of these factors within a unified system improves both predictive capability and analytical clarity. The proposed system provides reliable deficiency risk classification and supports data-driven nutritional assessment.

Despite its strengths, the study acknowledges certain limitations, including reliance on user-reported dietary data and the absence of automated intake tracking mechanisms. These limitations highlight opportunities for future work, such as the integration of wearable devices, real-time food tracking technologies, and computer vision-based food recognition systems. Advanced extensions, including adaptive personalization and prioritization mechanisms, can also be explored to further enhance system functionality.

In conclusion, this research demonstrates that a data-driven approach combining machine learning and nutritional analysis can effectively support predictive dietary assessment. The proposed system provides a practical and scalable framework for analyzing dietary patterns and identifying potential nutritional deficiencies. This work contributes to the advancement of intelligent nutritional systems and establishes a foundation for future developments in digital health technologies.

6. DESCRIPTION OF PERSONNEL AND FACILITIES

6.1 Supervisor

Mrs. Lokesha Weerasinghe

Senior Lecturer

Faculty of Computing | Information Technology

Sri Lanka Institute of Information Technology

Malabe

6.2 Co-Supervisor

Mrs. Chathurya Kumarapperuma

Assistant Lecturer

Faculty of Computing | Information Technology

Sri Lanka Institute of Information Technology

Malabe

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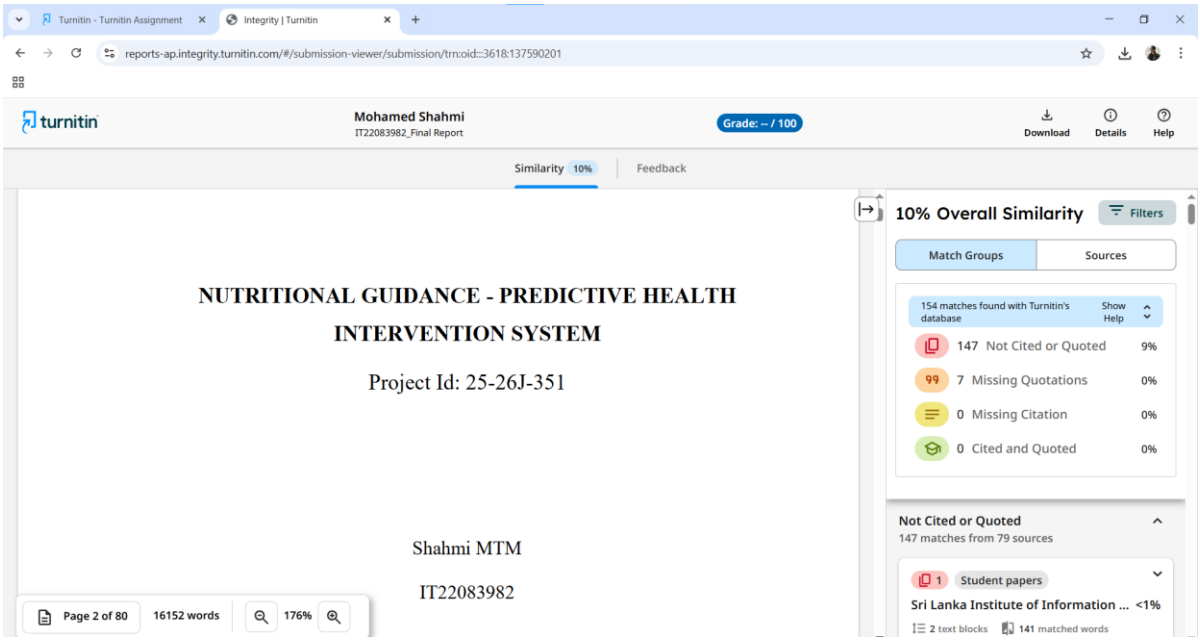
8. APPENDICES



Appendix 1 -- Plagiarism Report